

The Shadow Files

Volume 2
Attribution

YEHOR SELIN

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Before You Begin

This is the second volume of The Shadow Files — a practical course in open-source intelligence, OSINT. It carries straight on from Volume 1. If you have not read the foundations — the loop, the confidence ladder, protecting yourself — read them first. This book assumes them.

Everything here is invented

Every name, username, email address, phone number, photograph, and case study in this book is entirely fictional. Each one was built to teach a method — never to describe a real person. Any resemblance to a real individual, account, or event is pure coincidence.

Because every detail is invented, the examples are shown in full — complete, ordinary-looking data instead of blanked-out symbols — so you see exactly how a real trail behaves across platforms without any of it touching a real person.

The rules, in one line

Open sources only. Nothing that needs a login or a password. Respect people's privacy and the law. That is the whole frame, and it does not loosen as your skill grows.

Two things worth keeping in mind. Know the law where you work — a client hiring you to investigate someone is not the same as the law allowing you to collect everything about everyone near them. And most of the damage in this craft is done to people who were never the subject, so take only what answers your question and stop there.

A professional is not the person who can find the most. It is the person who knows where to stop.

One Clue Was Never the Point

What changes in Volume Two

In Volume 1 you learned to pull one thread. You took a single clue — a username, an email, a number — and, step by step, followed it to a person. That is the foundation, and everything here stands on it.

But real people do not live on one platform. They live scattered — a photo here, a comment there, an old account they forgot, a relative who tags them, a leaked email that ties two names together. One thread is where you *start*. It is almost never where you *finish*.

Volume 2 is about the finish. It is about taking threads from many places and weaving them into one confident identity — and doing it under real pressure: private profiles, look-alikes, deleted content, and your own eagerness to be right.

Who this book is for

You, if you have the foundations and want to go from *finding a clue* to *building a case*. You do not need to be technical. You need the mindset from Volume 1 and the patience to work a person across half a dozen platforms without losing your discipline. And if a technical word ever trips you up, there is a glossary at the back — flip to it any time.

What you will be able to do by the end

Follow one person across social networks and messengers. Read what a photo and a face can and cannot tell you. Use leaked data to connect identifiers — never to break in. Draw a relationship map and let it lead you. And, hardest of all, judge *how sure you are*.

What this volume is not

It is not a tool manual, and it is not the whole craft. Deleted-account reconstruction, geolocation, and infrastructure wait for Volume 3; the professional's world of automation, due diligence, and counter-investigation waits its turn after that. Here we master one thing well: **attribution across platforms**.

One clue opens a door. This book is about the rooms behind it.

How Volume Two Works

The series, this book, and how each chapter is built.

The Shadow Files is a series of four volumes. Each one takes you a level higher, and each assumes only what the last one taught.

- **Volume 1 — Foundations.** The mindset, the rules, and your first real skills.
- **Volume 2 — Cross-Platform Attribution.** Following one person across social networks, messengers, photos, and leaks — into a single, confident identity. (*This book.*)
- **Volume 3 — Hard Targets.** Deleted accounts, geolocation, infrastructure, and the tricks that surprise people.
- **Volume 4 — The Professional.** Corporate work, reports, counter-investigation, and turning skill into a craft.

A note on scope: we are still after people, not servers, domains, or physical locations — that is real OSINT too, but a different toolkit for another day.

Inside this volume

1. Private Is Not Closed 2. The Username Web 3. Telegram, Up Close 4. Faces and Photographs 5. Leaks, Handled Cleanly 6. The Identity Graph 7. Making the Signals Agree 8. Fighting Your Own Certainty 9. The Full Attribution

How every chapter is built

The same rhythm as Volume 1, so you always know where you are:

- The idea — in plain words, no jargon.
- The method — how a professional actually thinks and works, step by step.
- A worked case — a short, invented investigation you can follow, with the dead ends left in.
- An exercise — something safe and legal you do yourself.
- A recap — the few things to remember.

One promise about tools

As before: method first, tools second. Some tools named here may be gone by the time you read this, and new ones will have taken their place. That is fine. The *thinking* does not change. Learn the method, and any tool is just a faster way to do something you already understand.

A Bridge from Volume One

The foundations, in one page, before we go deeper.

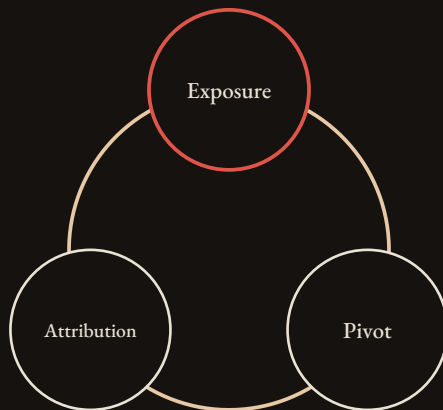
Before we go anywhere new, let me put the foundations back in your hand — quickly. If any of this feels unfamiliar, that is your sign to go back to Volume 1 before you go forward here. Everything in this book leans on these four things.

The loop

Every investigation, however big, is the same three moves, run in a circle.

Exposure → pivot → attribution.

THE LOOP



one clue, run in a circle until it points at a person

The three moves run as a cycle — attribution feeds the next exposure.

Exposure is a trace a person left in the open. Pivot is turning that trace into the next one. Attribution is when enough traces line up to point at one real identity. You start from one clue and run the loop until the picture is clear. Volume 2 does not replace this loop. It runs it across many platforms at once.

The confidence ladder

Because a match is not a fact, we always say *how sure* we are, in four fixed words:

- **weak** — one unconfirmed clue. A lead, never a conclusion.
- **moderate** — two independent clues agree.
- **strong** — several independent signals that are, together, hard to explain by any rival hypothesis. Usually that means three or more. But two heavy, diagnostic ones (say a byte-identical avatar file plus a verified self-link) can reach strong, and three weak, common ones may fall short. Count is a rule of thumb; how hard the agreement is to explain away is the real test.
- **confirmed** — a hard unique identifier or a genuine self-link, independently nailed down so anyone could re-check it.

THE CONFIDENCE LADDER



Four rungs you climb only on independent signals: weak, moderate, strong, confirmed.

You will meet these four words in two places: grading a *single signal's* strength, and grading your *overall conclusion*. Same words, linked by the rule above — a lone strong signal is not yet a strong conclusion.

You will use this ladder harder in Volume 2 than anywhere, because attribution across platforms is exactly the place where eager beginners turn a pile of *weak* into a false *confirmed*.

Same handle is not same person

The single sentence to carry out of Volume 1: a matching name, handle, or face is a *lead*, not proof. Two strangers can share a username. A common face can mislead. You confirm with independent signals — content, dates, connections — not with the match itself.

The line, and where to stop

Open sources only. No logging in, no pretending to be someone else, no getting past a lock. And the rule above every technique: the one who knows where to stop, not the one who finds the most. The methods ahead make it easy to keep going — so take only what answers your question, and stop there. Before the first pivot, be able to say in one plain sentence why *this* person must be looked into, and for what legitimate question. If you can't, you have curiosity, not a case.

Foundations in hand? Good. Let us go across.



CHAPTER ONE

Private Is Not Closed

A locked profile is a wall you walk around, not a door you force.

You click the profile. The avatar loads. The name loads. Then nothing — no posts, no friends, no photos. A padlock where the life should be. Most people stop here. They write "private, dead end" in their notes and move on.

I understand why. A locked profile feels like a closed door, and closed doors are meant to be respected. The person behind it is someone — a mother, an engineer, a widower, a kid trying to grow up in peace — who reached for a setting and thought that was the end of it. Hold onto that as you read this chapter. The

work below is not about defeating them; it's about noticing what the world around them already said out loud.

Private controls what they publish. It almost never controls what everyone else did.

A locked account restricts your direct view of one page. It does not restrict the friend who tagged them at a wedding. It does not restrict the group they joined in 2019 that still lists members. It does not restrict the archive that snapshotted the page before it went private. And often — though no longer always — it does not hide the profile picture, which many platforms keep visible so the account stays findable at all.

One thing to fix in your head before we start. This chapter's job is narrow: you already *suspect* an identity — a name, a face, a claim — and you want to know whether a locked profile belongs to that person. You are here to **confirm a match**, using the open people and records around the target — not to discover an identity you do not yet have. Finding a hidden *second* person — an unknown relative, an alias no one handed you — is the work of Chapter Six, which walks the family ring rather than the friend ring.

The idea: enumerate the open ring, not the closed centre

Picture the target as a dot. Around that dot is a ring of other people and other records — friends, taggers, commenters, group co-members, old cached pages. The centre is dark. The ring is lit.

ACCESS LAYERS



the centre is shut —
the rings around it
are wide open.

Work inward from
what others left
public.

The locked centre, ringed by the layers that stay open: public posts, friends-of-friends, caches and archives, and old leaked copies. You read the rings, not the centre.

Your job is not to break into the centre. Your job is to walk the ring and read everything the ring left facing outward, then assemble those fragments into a picture of the centre.

This reframes "private" completely. A private profile is not an absence of information. It is information that has been *displaced* — pushed off the target's own page and onto other people's

pages, into archives, into caches. You just collect it from a different address.

There are seven common leak layers. Learn them as a checklist — but read the next section first: some of these have decayed hard since the mid-2010s, and you need to know which moves are still live before you reach for one.

- **Mutual connections and "people you may know."** Platforms suggest connections. Those suggestions map the target's graph even when their friends list is hidden. (This signal runs both ways — mind the warning below.)
- **Tags and mentions by others.** A private user tagged in a public friend's post is fully visible — on the friend's page.
- **Comments.** The target comments on public pages, public news posts, public groups. Those comments carry their name and often their photo.
- **Group and page memberships.** Membership lists, event attendee lists, and "likes" often survive privacy settings because they live on the group's side.
- **Cached and archived copies.** Search-engine caches and web archives may hold a version of the page from before it was locked.
- **The profile picture and banner.** Often public, so the account stays findable — but this is now platform-dependent, and some services blur or gate it. When

it *is* open, it is a clean face for reverse search and a thread to an alternate account under a different name.

- **Friend-of-a-friend (FOAF) overlap.** The target is closed, but some of their friends are open. Someone who recurs across several of those open lists — and matches the face and name — is very likely inside the target's cluster. (Reviewed by hand; see Step 6.)

Notice what none of these require: no login to the target's account, no password guessing, no deception. You read what other people already published in public. That is the whole game.

What still works in 2026, and what quietly died

Older OSINT guides — including parts of this book's first edition — leaned on a handful of clever platform tricks. Several of them no longer work the way they once did, and nothing dates an investigator faster than confidently running a move the platform closed years ago. So learn the idea under each trick, and find the current equivalent for the platform in front of you.

Three moves in particular have decayed:

- **Grabbing the numeric ID from logged-out page source.** For years you could open a profile's raw HTML, search for `id` or `profile_id`, and read the permanent numeric ID straight out of the page. On the major platforms this is now largely closed — the page is a JavaScript shell and the ID is not sitting

in the source anymore. **The principle holds:** every account still has a stable numeric ID behind its handle, and that ID still survives renames. You just have to reach it through whatever the platform exposes today — a share-link redirect, an embed, an official lookup — rather than a view-source grab. Treat "view source and read the id" as an illustration of the idea, not a step you can count on.

- **The two-numeric-ID mutual-friends URL.** There used to be a URL you could hand two numeric IDs and get back the people they share. Facebook Graph Search, which powered much of this, was removed in 2019, and that mutual-friends endpoint went with it on the major platforms. **The principle holds:** shared connections still map a hidden graph, and some platforms still show mutuals in the interface. But the tidy paste-two-IDs URL is gone — do not teach it or lean on it as a live step.

- **Scraping and intersecting friend lists (FOAF).** The classic move was to pull several friends' full friend lists automatically and intersect them. Two problems in 2026. First, friends lists are hideable by default now and frequently hidden, so there is often nothing to pull. Second — and this matters everywhere else in this book — automated friend-list scraping breaches the platform's Terms of Service. We do not do it. **The principle holds:** the target reappears inside a known cluster of open people. But you reach it by *manual, open review* of the lists people have chosen to leave public, not by scripting a

harvest.

Before you look: don't light yourself up

Here is a mistake that even experienced people make, and it is new enough to Volume 2 that it deserves its own stop sign.

When you view a target while logged in to a research or "sock" account — a clean, separate identity you keep only for investigating, never your real profile — you are not invisible.

The platform's "people you may know" machinery is **bidirectional** — the same signals that surface the target to you can surface *your sock account to the target*. Repeatedly opening someone's profile, letting the app read your contacts, or sharing a network with them can push your investigative account into *their* suggestions. That tips them off, burns the account, and in a sensitive matter can compromise the whole engagement.

So set up before you look. Use a clean, well-worn research identity that is not linked to your contacts, your phone's address book, or your real accounts. Disable contact syncing. Prefer logged-out and archived views wherever they still return what you need. Assume that anything you do while logged in can echo back to the person you are watching, and act like it.

The method, step by step

Step 1 — Lock the anchor. Before anything, resolve the one thing that does not change: the stable numeric ID behind the vanity name. Vanity handles change; numeric IDs are permanent, and every technique that follows pivots off that anchor. Reach it through whatever the platform exposes today, and write it down. It survives the target renaming themselves next week.

Try it (2026). Pick one concrete route to turn the name into the number. On Telegram, forward one of the account's messages to an ID bot (a @username_to_id-style resolver — e.g. @username_to_id_bot or @usinfobot) and read back the numeric user ID. On Instagram, open a logged-out share-link like [instagram.com/<handle>/](https://www.instagram.com/<handle>/) in a clean browser and read the numeric id out of the page URL or a shared post's URL; on Facebook or X, an embed of one of the profile's posts exposes the same number the same way. On most platforms, a public "username to ID" web lookup does the same job from logged-out. Whichever you use, you are looking for the same thing: the number that lives under the name. Caveat: the specific bot, endpoint, or lookup site named above may well be dead by the time you read this — the shape lasts (find the number under the name), the exact door does not.

Step 2 — Harvest the picture. If the avatar is public, pull the full-resolution version — many platforms still serve a large

image even when the wall is closed. Save it. You will reverse-search it later, because the same face under a different, *unlocked* handle is one of the most common ways a private profile falls open. If the picture is gated or blurred, note that and move on; do not treat a missing avatar as a missing case.

Step 3 — Map shared connections. If you already have one known associate — an "orbit" profile such as a spouse, a colleague, a sibling — look for connections they share with your target. Some platforms still surface mutuals in the interface; the old paste-two-IDs URL does not. Do this from a properly prepared research account (see the warning above), and remember what a shared connection proves: a *link*, not an *identity*. You still have to look at the human on the other end.

****Step 4 — Search *around* the target, not *on* them.**** Use the platform's own search box with the target's name plus context words: a surname, an employer, a town, "birthday." This surfaces content on *other people's* pages that the target's own timeline suppresses. Then leave the platform entirely and dork the open web — `site:platform.com "their name"`, the numeric ID, any unique phrase from the bio.

Step 5 — Walk the archives. Feed the profile URL to a web archive and a search-engine cache. If the page was ever public, a snapshot may exist. Walk the calendar backwards — *older* snapshots are often richer, taken before the person tightened up. Check what the snapshot actually captured, though: a snapshot

taken *after* the lock preserves only the wall.

Step 6 — Review the open cluster (FOAF, by hand). Take three or four confirmed friends whose own friend lists they have left public. Read those lists yourself and note who recurs. A person who appears across several of your target's friends, and who matches the avatar and name, is very likely inside your target's cluster. Manual review only — no automated pulling, no scripted intersection.

Step 7 — Rank every fragment on the ladder. A shared first name from a tag is weak. A tag *plus* a matching face *plus* a shared connection is strong. Weight, then decide.

A worked case, dead ends and all

An investigative newsroom asked me to confirm one thing before they ran a story. A man calling himself Rui Costa had given an on-the-record quote to another outlet, presenting himself as the site foreman and safety spokesman for a troubled construction project — and a fully private Instagram account, handle `r.costa.builds`, avatar of a man at a building site, no posts or followers visible, was being passed around as proof he actually worked there. Did that locked account belong to the same Rui Costa the paper was about to quote as an authority? Fictional names, real method.

Anchor first — and straight into a dead trick. Out of habit I went for the logged-out page source to lift the numeric ID.

Nothing. The page was a JavaScript shell.

```
Logged-out page source
-----
view-source: r.costa.builds
search "profile_id" ... 0 hits
search "user_id"    ... 0 hits
page is a JS shell, no id in HTML
-----
2026: this grab is closed here
```

The old numeric-ID trick, dead on arrival.

So I did what the principle demands: I stopped mourning the trick and reached the stable ID through a share-link redirect instead. Anchor secured, just by a different door.

The avatar. The picture was public, so I pulled the full-res site photo and ran it through two reverse-image engines and one face engine.

```
Reverse image + face search
-----
QUERY r_costa_avatar.jpg
engine A 0 visual matches
engine B 1 match
  - stockphotos "man at job site"
face engine 1 candidate 68%
  - linkedin "Rui Costa", builder
-----
stock hit = generic, drop it
```

One stock trap, one thin face lead.

The stock-photo match was a generic "man at a job site" image re-used everywhere — a shared stock picture links nobody.

Weak, rejected. The face-engine candidate, a LinkedIn "Rui Costa" listed as a builder, was thin but real — a 68% face score, face only, no identical-file match. I parked it.

The archive dead end. The private profile had a clean vanity URL, so I checked a web archive for an older, public version. Two snapshots existed — promising, until I opened them.

```
Web archive lookup
-----
URL  instagram/r.costa.builds
snapshots found: 2
    2020-08-02  captured
    2023-01-19  captured
opening 2020-08-02 ...
    login wall preserved
opening 2023-01-19 ...
    login wall preserved
-----
both = platform login gate
```

Two snapshots, both walls. Nothing.

Both snapshots had captured only the platform's login gate, not a public version of the page. An archive keeps what was visible at capture time — here that was a wall. Dead end. Note it and move on; not every layer pays out, and a run where every layer pays out is usually one you should distrust.

What actually held. Two threads survived. First, the LinkedIn "Rui Costa" — same face as the avatar to the eye, 68% by the engine — listed the exact contractor the quote had named as his employer. Second, searching the target's name plus that

employer surfaced a public Facebook post by a friend, "Tiago M.," who had tagged a Rui Costa at a job site in 2023 on the same project the story described — content displaced clean off the private profile and onto a friend's page.

Where that leaves the ladder. Line them up honestly. The face link is moderate — a decent likeness, face-only, 68%, no identical-file match to nail it. The public tag is moderate too: it ties *a* Rui Costa to the project, but not conclusively to *this* private Instagram. Two moderate threads pointing the same way is a real signal, but two moderates are not a strong one, and stacking them does not promote them.

So I reported it as moderate — likely the same person, but one independent thread short of strong, and said exactly that. I told the newsroom what a third thread would look like — an identical avatar file elsewhere, a mutual who could place the accounts together, a bio detail matching across platforms — and that until one landed, I would not sign off on strong. A quote that hangs on identity is not one you run on a moderate. The editors got an honest rung, not a flattering one. That restraint is the job.

Where to stop

That profile was private *on purpose*, and everything I used was published by someone else in public. That is the line, and it is a real one. Reading a friend's public tag is open-source work. Talking that friend into accepting a fake request so you can see the private list is not — it is deception, it breaks the platform's rules, and in a legal matter it can poison the whole case. Look through the windows other people left open; do not pick the lock.

And respect the *original context*. A photo a friend posted at a party was meant as a party photo, not as evidence in your file. Collect what serves a genuine public interest, and when that interest runs out, stop.

Exercise

Do this on yourself. Log out of everything, open a clean browser, and try to rebuild your own private profile from the ring only — never from your own page.

- Find your stable numeric ID through whatever route your platform still exposes. Write it down.
- Reverse-search your own avatar, if it is public. Where else does that face or file appear?
- Search your own name plus your town on each platform's search box. What did friends tag you in?

- Check a web archive for old public versions of your profile — and check what each snapshot actually captured.
- Read three friends' public lists by hand. Do you reappear across them?

Then write your own confidence line: from the ring alone, could a stranger reach *strong* on you — or only *moderate*? Most people are shocked at how far the ring carries. Fixing your own ring is the best privacy lesson there is, and the fastest way to understand the method from the inside.

Recap

- Private restricts your view of one page; it rarely restricts what *others* published about the target.
- This chapter's job is to confirm a suspected match through the open ring.
- Map the open ring around the target — shared connections, tags, comments, groups, caches, archives, the avatar, hand-reviewed FOAF — not the closed centre.
- Anchor on the stable numeric ID first; it survives renames.
- Know which classic move died: page-source ID grabs, two-ID mutual URLs, and scripted friend-list scraping are closed or against the rules. The principle survives each one; find the current equivalent.
- Don't light yourself up: viewing from a logged-in sock account can surface *you* to the target through "people you

may know." Prepare the account first.

- Expect dead ends: dead tricks, stale archive snapshots, generic shared images, same-name-different-human matches. Reject them out loud.
- Two moderate threads are not a strong one.
- Read open windows, never pick locks; respect the original context; stop when the public interest does.



CHAPTER TWO

The Username Web

One handle is a thread. At this level you weave the whole web.

In Volume 1 you learned to take a handle and check whether it exists on other sites. That is the base move — namechecking. Useful, and not enough. It hands you a list of URLs that responded, and a list of URLs is not an identity.

This chapter is about the next altitude. Not "does this string exist elsewhere" but "which of these hits are the *same human*, how does this particular person *mutate* their handles, and where does the whole web of them collapse into one defensible identity."

A handle is a lead. A person is a conclusion. The work is the distance between them.

The idea: strings are cheap, people are expensive

Enumeration is cheap. You can run one handle against seven hundred sites in a minute and get forty green ticks. That feels like progress. It is not — it is forty *hypotheses*, most of them wrong. Some are reserved-name placeholders. Some are unrelated humans who happened to like the same word. A few are your target.

At this level you invert the effort. Enumeration is the cheap base of the pyramid; the expensive top is confirming that a specific hit is a specific person by their *content, dates, and photos* — never by the string alone.

Three ideas do the heavy lifting here.

- **Naming grammar.** This is the chapter's core idea, so slow down on it. A person does not pick handles at random. They mutate them in a consistent, personal way — always an underscore, always a trailing birth year, always the same lucky number, always the same leetspeak swaps, the same beloved stem recombined. That habit is a *fingerprint made of choices*. Two people can share a string by accident; almost no two people share a whole grammar by accident. So the payoff runs in two directions. Backward, a matching grammar corroborates that scattered accounts share one author.

Forward — and this is the part namechecking never gives you — once you have read the grammar you can *generate handles the person has not shown you and go find them*, predicting where the target lives instead of reacting to a tool's green ticks.

- **The homonym trap at scale.** A common handle is many humans. shadowwolf is not a person; it is a demographic. The rarer the string, the higher the odds that two hits share an owner. Rarity is your friend; commonness is a swamp.
- **Confirmation by convergence.** One green tick is weak. A green tick *plus* the same rare avatar *plus* a bio phrase reused verbatim *plus* creation dates that fit one life *plus* the same social graph — that is how a string becomes a person. Volume 1's ladder, applied at scale.

The method, step by step

Step 1 — Enumerate wide, with more than one tool. Run the seed handle through at least two enumerators. Each ships a *different* site list, so their results diverge — one finds a forum the other never checks. Two tools, more coverage, and disagreement between them is itself information.

Here is what a real run looks like — messy, not clean:

```
SEED  nyx_falken
-----
[+] github      200  live profile
[+] reddit      200  live profile
[+] steam       200  live profile
[-] instagram   404  not found
[+] twitter/x   200  live profile
[?] tiktok      timeout (retry)
[?] mastodon    429  rate-limited
[+] pinterest   200  RESERVED page
[-] twitch      404  not found
-----
9 checked  5 live  2 404  2 unknown
NOTE: pinterest = placeholder, not
      a real profile. verify by eye.
```

What a username enumerator prints

Read that box the way a professional does. Five live hits — but the Pinterest "hit" is a reserved-name placeholder, a false positive the tool cannot tell from a real profile. The TikTok timeout and the Mastodon rate-limit are *unknowns*, not "no" — a failed check usually means the network blocked you, not that the account is absent. Re-run those later from a different address before you write them off. The tool tells you a URL responded — never who is behind it.

Step 2 — Triage every hit by eye. Open each live profile. Is it your subject, an unrelated human, or a squatter? Harvest pivot material from the real ones: display name, bio, linked URLs, other handles they mention, avatar. Every new handle they *declare themselves* is gold — a self-declared link between two accounts is near-confirmation, far stronger than any string

match you inferred.

Step 3 — Read the naming grammar. Lay the confirmed handles side by side and find the pattern. Does this person always append a birth year? Always swap a for 4? Always run first-name-underscore-noun? Once you see the grammar, generate the variants they *haven't* used yet and test those. You stop guessing randomly and start predicting.

Step 4 — Weight by rarity. Before you trust any cross-platform match, ask how rare the string is. A distinctive coined word — `nyx_falken` — carried across four platforms is a strong prior. A dictionary word — `nightowl` — carried across four platforms is almost meaningless, because thousands of unrelated people picked it independently. Adjust your confidence to the string's rarity *before* you look at anything else.

Step 5 — Confirm on content, not the string. This is the step amateurs skip. For each candidate, line up the *evidence that survives a name*: the avatar (same file, or same face under different photos), bio phrasing reused word-for-word, creation dates that fit one plausible timeline, the same friends appearing across accounts, the same writing tics. When several of these converge on one person, you have an identity. When the *only* thing shared is the handle, you have a coincidence.

Step 6 — Place it on the ladder and stop at the honest rung. Report what you actually reached — not the rung you wish you

had.

Try it (2026). Run one handle through Sherlock or Maigret — or a WhatsMyName-style web checker — to get a live cross-platform hit list, then open each hit and verify it by content, never by the green tick alone. Site lists and detection rules rot, so treat any single tool's output as a starting point and re-check misses from a fresh address.

A worked case, with a look-alike rejected

The seed was one handle from a marketplace scam report: `nyx_falken`. Fictional throughout, real method. The client wanted a single identity behind it, or an honest "cannot get there."

Wide enumeration gave me the run in the box above — GitHub, Reddit, Steam, Twitter/X live; Instagram and Twitch absent; TikTok and Mastodon unknown; Pinterest a placeholder. Five real doors, and `nyx_falken` is a coined word, not a dictionary term. Good rarity prior. I started hopeful but disciplined.

Triage. GitHub was the richest. The profile bio read: *"code by night, ramen by day — falkennest.dev."* It self-declared a personal domain. The Reddit account posted heavily in a mechanical-keyboards subreddit. Steam showed a public profile, cyberpunk avatar. Twitter/X: same coined handle, an art-focused account.

The naming grammar. GitHub, Reddit, Steam, and the personal domain all used `nyx_falken` or `falken`. The domain — `falkennest.dev` — added a new stem, `falkennest`. I re-enumerated that new stem on a fresh pass — the earlier run had only ever tested `nyx_falken`, never `falkennest` — and it turned up a Mastodon account `@falkennest@fosstodon.org`. The grammar was now legible: `nyx` prefix or `falken(nest)` stem, lowercase, occasional underscore. That let me predict and confirm the Mastodon account rather than stumble on it.

The look-alike. The Twitter/X `nyx_falken` looked like an easy fifth platform. I nearly logged it as a match on the handle alone. Then I checked content instead of string.

```
HANDLE  nyx_falken  (twitter/x)
-----
display  "Nyx | comm status: OPEN"
bio      furry artist, she/her, EST
avatar   original art, NO reverse hit
created  2019-06 (decoded from id)
-----
vs GitHub nyx_falken
   joined 2022-11 (page, not id)
   bio    male, ramen, keyboards
   domain falkennest.dev  (unlinked)
-----
VERDICT  same string, different human
```

What a cross-platform check surfaces

Everything diverged. Different pronouns, different self-description, an account created *three years earlier* than the GitHub one — and note the two dates come from different

sources: the Twitter/X date is decoded from the account's snowflake ID — a number with the account's creation time mathematically baked into it, so it decodes to an exact date — while the GitHub date is just the "Joined" line printed on the public profile page, not decoded from any ID — an original avatar with no shared file, and no link back to falkennest.dev. Same rare string, two unrelated humans — the homonym trap at scale, which strikes even coined words because a second person can independently coin the same thing or grab a freed handle. Rejected as the target. I logged the Twitter/X account as a separate person and struck it from the identity.

Confirmation by convergence. What actually held together: GitHub, Reddit, Steam, the falkennest.dev domain, and the Mastodon account. I wanted a hard identifier off that domain — an email, a registrant name — and I did not get one, and it is worth saying why, because a lot of older tutorials will lie to you here. In 2026 a .dev WHOIS record is privacy-redacted by default. The registrant name and email are hidden behind the registrar; there is no public page with an email local-part to read off. That easy trick is gone for most modern TLDs. Two narrow avenues remain, both with caveats. *Historical* WHOIS — snapshots archived before privacy became the norm — sometimes preserves a raw registrant, but this domain was far too young to have any. *Certificate-transparency logs* record every TLS certificate issued for a domain and occasionally leak a subdomain or a contact address; here they showed only the

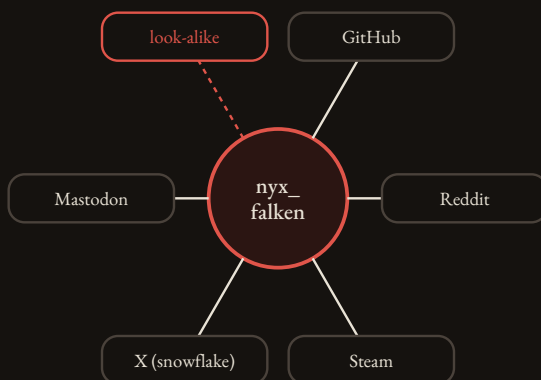
registrar's own. So the domain earned its weight the honest way — as a link the subject *declared themselves* in the GitHub bio, not as a secret email I decoded.

The rest of the set held on evidence that survives a rename. The Mastodon bio reused a verbatim phrase from GitHub — “*code by night, ramen by day*” — word for word, an unusual line unlikely to be coincidence. Steam and Reddit shared the same cyberpunk avatar file (identical asset, not a look-alike). Creation dates lined up into one plausible 2021–2022 arc.

Signals converging: verbatim bio phrase across two platforms (strong), identical avatar file across two more (strong), self-declared personal domain linking the set (strong), consistent naming grammar (moderate). Five platforms, one person — and crucially, *not* the Twitter/X account that shared the handle.

I reported strong — one identity across GitHub, Reddit, Steam, Mastodon, and the personal site — and named the rejected Twitter/X look-alike explicitly so nobody downstream re-added it. I did not claim confirmed: the subject never self-linked every account in one place, and no single hard identifier resolved the whole set uniquely. Strong was the honest rung.

THE USERNAME WEB



one habit reused across platforms — verify each by content

One seed handle fans out to many bits; content and grammar pull the real ones into one identity while a same-string look-alike is cut loose.

A note on the traps you will meet

Recycled handles are the classic snare. Someone abandons `nyx_falken`; a year later a different person grabs the freed name. Now the string spans two humans across *time*, not just space. Creation dates are your defence — a handle that "predates" the person's known activity is a rename or a recycle, not a time machine.

Generational nicks and leetspeak widen the grammar. A person who was `xX_dragon_Xx` at fourteen may be `dragon.eth` at twenty-eight — same human, evolved naming. Read the drift, don't be fooled by it. And remember the LLM-era wrinkle:

writing style, once a reliable tie-breaker, is being flattened by AI-assisted composition. Lean harder on hard invariants — creation dates, shared image files, self-declared links — when the prose gets suspiciously smooth.

Exercise

Pick a well-known open-source developer with a public presence — someone who *wants* to be found across platforms, so this stays firmly in open territory. Do not pick a private individual.

- Seed their most-used handle into two different enumerators and diff the results.
- Write out their naming grammar in one sentence. Predict one handle they use that neither tool found, then verify it.
- For three hits, confirm same-person on *content* — avatar file, bio phrasing, creation date — not on the string.
- Deliberately hunt for a look-alike: someone sharing the handle who is *not* your subject. Reject them in writing, with your reason.
- State your final rung and the one piece of evidence that would push you to the next.

Recap

- Enumeration is cheap and produces hypotheses, not identities; the expensive, decisive work is confirmation.

- Read the person's naming grammar — their consistent mutation pattern — and predict handles rather than only stumbling on them.
- Weight every match by rarity: coined strings carry attribution weight, dictionary words are a swamp.
- The homonym trap strikes even rare handles — same string, different human, or a recycled name across time. Reject look-alikes explicitly.
- Confirm on what survives a rename: avatar file, verbatim bio phrasing, creation dates, social graph, self-declared links — never the string alone.
- Treat 404 as "no," but timeout and rate-limit as "unknown" — re-check before writing an account off.
- Place the whole web on the ladder and report the rung you reached; strong is honest, confirmed needs a self-link or a unique hard identifier.



CHAPTER THREE

Telegram, Up Close

Why the numeric ID is the anchor, and the handle is just a costume

People treat Telegram as a private room. It is not. It is a glass box they mistake for a wall.

They lock down their profile, hide their phone number, pick an anonymous handle — and then they go and *post* in a dozen public groups, forward each other's messages, and log in at the same hour every night. Every one of those is a trace left in the open. None of it needs a warrant, a login to their account, or a single message sent to them.

This chapter is about reading those traces well. Here we go deep on one platform, and on the one fact that makes Telegram attribution durable.

One 2026 note before we start, because it changed the stakes. After Telegram's September 2024 policy shift — the one that followed its founder's arrest in France — Telegram will now disclose a user's IP address and phone number to authorities under a valid legal request in a criminal case. That is real, and it matters. But it is a lever for law enforcement with a court order behind them, not a tool for you, and nothing in this chapter touches it. What the shift confirms from the outside is exactly the lesson we build on: the account's numeric ID is the durable spine of the identity, the thing every system keys off. Usernames went the other way — they are now openly traded as assets on Telegram's own Fragment marketplace, bought and sold and re-registered to strangers. The costume has a resale value; the number does not change hands.

The handle is a costume. The number underneath it is the person.

The one thing everything hangs on: ID, not username

Every Telegram account carries a permanent numeric ID, handed out at registration and never changed. The @username sitting on top of it is mutable — it can be renamed, dropped, sold, or claimed later by a total stranger.

That gap is the whole game.

- The numeric ID is a positive integer, assigned roughly in order over time. Older accounts have lower numbers. It is the anchor.
- A group or channel ID is negative — a supergroup shows as -100 glued to its internal number, while a plain (non-super) group is just a negative number with no -100 prefix (e.g. -4829103), a small tell that the chat is older or simpler.
- The @handle is a snapshot. It tells you who holds that name *today*. Tomorrow is not promised.

So the first move in any Telegram case is the same, every time: resolve the handle to its numeric ID, and from then on treat the *ID* as your subject. Everything durable — account age, past names, group history — keys off the number.

A resolver does the conversion. Some are bots (@username_to_id-style), some are no-login web tools. For a contact of your own, forwarding one of their messages to an info-bot returns the sender's ID — but only if they haven't set

forwarded messages to unlink from their account, in which case you get a name and no clickable profile at all. That is your first dead end, and it will not be your last.

```
resolve @nightferry_kv
id:      512883471
username: @nightferry_kv
name:    "kv"
status:  ok
```

A username-to-ID resolver in action

Write the ID down. It is the case anchor now.

Try it (2026). Take a @handle and resolve it to its numeric user ID: forward one of the account's messages to a @username_to_id-style ID bot — e.g. @username_to_id_bot or @usinfobot; try one, and if it errors or is gone, try another — or paste the handle into a no-login username-to-ID web tool, and read the number back. That number is permanent. But do it carefully: any third-party resolver or ID bot *sees your query* — your account, the time, and the exact ID you asked about — so run it only from a clean research account, never against the rare centre-of-case ID you most want to protect. A cheerful "helpful" bot is exactly what a honeypot looks like.

The tools, in plain words

Almost everything in this chapter leans on small community tools — mostly bots, a few web pages. Before we use any of them, hold one fact steady: the *type* of tool matters far more than its name, because whatever specific bot you find will be dead or renamed by the time you need it again. Learn the five categories.

- Resolvers turn a @handle into its numeric ID and back. This is the one durable trick, because the ID never changes.
- Age estimators guess when an account was created by reading its ID against a table of known signup dates. An estimate, never a real timestamp.
- History observers report the old names and handles an account once used — but only the changes the tool personally saw from inside shared groups.
- Membership indexers claim to list the public groups an ID belongs to, drawn from their own crawl of public chats. Always an undercount.
- Lookup bots take a phone number or a handle and hand back a crowd-sourced name. That is a rumour with a label on it, not an identity.

Why the number is durable: two quick reads off the ID

Once you hold the ID, two things fall out of it directly — no contact with the target needed.

Account age. Because IDs climb roughly in order, you can estimate when an account was born by interpolating between IDs whose real signup dates are known. Community tools keep tables of these anchor pairs and read your ID against the curve. It is an *estimate* — accurate where anchor points are dense, fuzzy in sparse regions — so treat it as " $\pm$ (days to weeks)", never an exact timestamp.

Why bother? It catches the lie of age. A "long-time member" whose account is three weeks old is a freshly-minted sock. A cluster of accounts all born the same afternoon is a coordinated batch. That is a *moderate* signal on its own, and a strong one when it lines up with other evidence.

Name and username history. Telegram publishes no history API. But observer bots have sat inside millions of public groups for years, quietly logging every name and handle change of accounts they shared a room with. Query one by ID and it reports the transitions it personally witnessed — the old personas, the rebrand from a dirtier handle to a clean one.

The hard caveat, and say it out loud: **you only see what the observer observed.** A rename that happened before that bot ever shared a group with the account was recorded by no one. Its absence from the report is not evidence it never happened. This is observation-bound intelligence — leads, not a complete ledger.

Finding a person through the groups they post in

Here is the move most beginners miss. You rarely need to break into anything. You need to find *where the person already speaks in public*, and read it.

Two directions:

Top-down — you have the person, you want their rooms.

Membership-index bots claim to list the public groups an ID is or was in. They work by having crawled public groups at scale, recording "(this ID) seen in (this chat)" tuples. You are querying *their* log, not Telegram. So: massive undercount, private groups never appear, "last seen" dates are lower bounds, and some of these bots are fed by scraped or breach data — stale, wrong, and sometimes unlawful to lean on. Use the output as a list of *places to go look by hand*, nothing more.

Bottom-up — you have the rooms, you want the person. If you already know several groups of interest, pull each member roster and intersect them. The account that shows up in *all* of them is your coordinating core — and you found it without trusting a single membership bot. A roster pull off a large supergroup caps at a few thousand names, so you get a sample, not the whole room; absence from your export never proves non-membership. Admins can also hide the member list entirely, and broadcast channels usually don't expose subscribers at all — you only see the people who comment.

Either way, the payoff is the same: once you're in a public group the target posts in, you read what they *say*, and saying reveals far more than membership.

Reading activity: routine as a fingerprint

A person's posting timestamps, gathered over a couple of weeks, draw a shape. Cluster them and the quiet stretch — the hours with no activity — is roughly when they sleep. That gap approximates their timezone, and timezone narrows region, even for an account with a fake name and no photo.

Layer on *what* they post: the language, the slang, the local references, the football club, the times they complain about their commute. None of this is one clean number. It is a slow accumulation of small tells, each weak, that together climb the ladder from *weak* to *moderate* and — with a corroborating hit from another axis — to *strong*.

Watch the traps. Scheduled posts lie about the poster's hours. A shared account run by two people in two timezones smears the pattern into noise. And a careful operator posts deliberately off-hours to mislead you. Timezone-from-timing is a lead you confirm, never a verdict you announce. Chapter Seven will show *why* this is only a lead: two different people from the same city and the same online scene share these patterns by default, so timing agreement between two accounts is not, on its own, evidence they're one person.

The operator sees your query

Here is the thing nobody selling these bots will tell you, and it is the most important paragraph in the chapter. Every time you send a query to a third-party resolver, history observer, membership indexer, or lookup bot, you are handing your question to a stranger. The person running that bot sees your querying account, the time you asked, and the exact ID or number you asked about. The tool is not a vending machine. It is a person in the middle, and they are reading the middle.

Think about what that leaks: your interest. Against a careless target it means nothing. Against an organized one — a fraud crew, a harassment ring, a hostile group with its own technical people — it can be the whole case. A real share of these "helpful" bots are honeypots: run by the very communities investigators look into, or by data brokers who log every querying account to resell. Feed one the rare ID at the centre of your case and you may have just told your target that someone is looking, and handed them a handle to look back at. This is, concretely, how an investigator gets made.

So work like the middle is hostile, because sometimes it is:

- Prefer the official client and your own tooling. Anything you can read by scrolling a public group yourself, read that way — it exposes nothing to anyone.

- Assume every third-party bot logs you. Query only from a clean research account that never links back to you or your client, never your real one.
- Don't feed the operator the story. Resolve an ID if you must, but avoid handing one bot the *whole* pattern of rare queries that would tell the person running it which case you are on.
- Against organized targets, treat resolver and lookup bots as burned by default. The convenience is never worth getting made.

Never run the ID-anchor or a cross-platform resolve to unmask someone who is anonymous for their safety.

An anti-corruption source, a protester under an authoritarian government, a whistleblower — the same handle-to-durable-number move that catches a harassment account is exactly what a hostile client will pay to point at them. A commission whose interest in unmasking that person is retaliation, not accountability, is a red flag to decline, not a brief to fulfil. The tools in this chapter do not know the difference. You do.

The lookup bots: passive is not undetectable

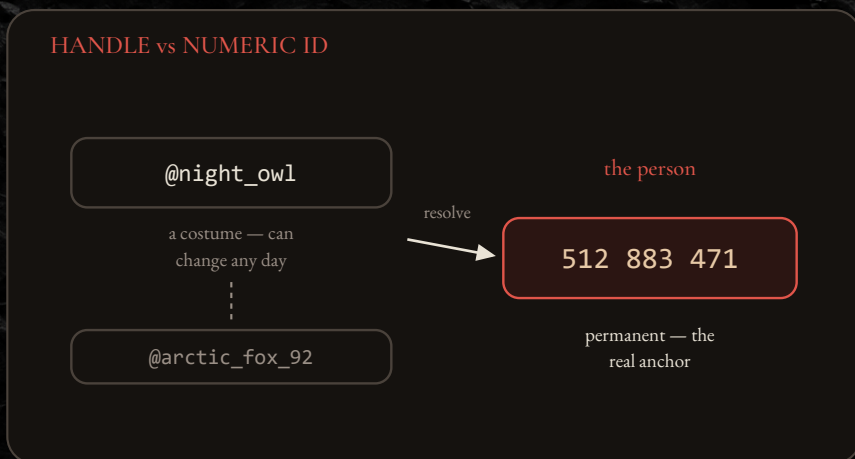
You will be tempted by the phone-and-username lookup bots — feed a number, get a name; feed a handle, get a "who is this". Use them with your eyes open, because they carry four honest problems.

- They are rate-limited. A free tier is often one lookup a day. Bulk querying gets your research account flagged and banned.
- They are sometimes wrong. A caller-ID or crowd-sourced name can be an *old owner* of a recycled number, or a label some stranger saved. It is a lead, not an identity.
- They sometimes trip terms of service — or are fed by breach dumps whose use may be unlawful under your mandate. Decide that *before* you query, not after.
- "Passive" is not "undetectable." A resolving account leaves footprints. Contact-sync against a real person's number is a data-protection act in many places. Adding a logging bot to someone else's group is not passive at all — it needs consent you don't have.

Worked case: from a clean handle to a confident name

I get a single handle from a client: @nightferry_kv, tied to a low-grade harassment campaign in a regional community chat. That is all. No name, no number.

Anchor first. Resolve it. ID 512883471. From here, the ID is my subject.



The mutable handle sits on top; the permanent numeric ID underneath is what account age, name history, and group traces all key off.

Age check. The creation estimate puts the account at roughly four years old. Not a fresh sock — an established identity. That is useful; it means there may be years of public trace behind it.

The first dead end. The profile itself is clean. No photo, name just "kv", bio empty, forwarded-message linking switched off. The account gives me nothing directly, and a lookup bot on the handle returns:

```
query: @nightferry_kv
name (crowd): "Andriy Petrenko"
phone:      not available
confidence: low / unverified
note: crowd label, source unknown
```


What a username lookup bot returns

A name. Tempting. And I *reject it* — for now. "Crowd label, source unknown" is exactly the caller-ID-style residue that names an old owner or a stranger's guess. It goes in the file as *unconfirmed, low confidence*, nothing more.

Pivot to the ID's history. I query a name-history observer by the numeric ID. It reports a former username the account cycled through two years back: @k_veranko. The current account is scrubbed; the *old* handle was not built to be.

Cross-platform on the old handle. I run @k_veranko — not the current one, the discarded one — through username enumeration. It surfaces a years-old, still-public account on another platform: same handle, an avatar, a bio linking a personal blog, and a first name plus initial: "Kateryna V."

Now the "Andriy Petrenko" from the crowd bot doesn't just look unconfirmed — it looks *wrong*. Two independent, self-published signals point to a different person entirely. The crowd label is discarded outright.

Corroborate on the group content. Back in the community chat, I read the target's public posts over two weeks. Quiet hours line up with a local sleep schedule — right region. Repeated references to a specific tram route and a named workplace. The old blog from the cross-platform hit mentions the same workplace.

That is two axes pointing the same way: a discarded handle traced across platforms to a blog persona, and public group content placing that persona in the same city and workplace. The avatar on the old account matches the blog; the workplace is named in two places she published herself.

Now look hard at what I actually hold, because this is where beginners over-claim. The name is a first name and an initial — "Kateryna V." — not a full identity. The blog could belong to a namesake. The shared workplace is suggestive, not decisive, in a city of that size. The numeric ID is a *confirmed* anchor for the *account* — but the jump from account to a specific named human still rests on soft, self-published matches and one recycled handle. Every link in that second chain is plausible. None of it is hard.

Verdict: I attribute to "Kateryna V." at **moderate** confidence — a solid, named lead, not a locked identity. To climb to strong I would want one more independent axis: a second platform tying the same face to the same full name, or a document. I have neither, and I will not manufacture them. So I report moderate, name the two supporting axes, flag the rejected crowd-name so the client sees the fork I chose, and write down plainly what would move the needle. The dead-end clean account was never the obstacle; the *old* handle it forgot to bury was the way in — it just did not carry me all the way home. Stopping at moderate, and saying so, is the finding.

Note what I never did: no private group under a false story, no message to the target, no logging link — every step read a public surface.

Exercise

Safe drill (your own account). Resolve your own handle to your numeric ID with any resolver. Then run the ID through a creation-date estimator and compare the result to the month you actually joined. Notice the error margin — that gap is exactly the uncertainty you must carry into every age claim you make about a *stranger's* account. Do not query other people's numbers for practice; use only yourself.

CTF-style (public, consenting data). Pick a well-known public Telegram channel with an active comment section. Without joining anything private: collect the visible posting timestamps of the three loudest commenters over a week and sketch each one's quiet-hours gap. Write down the timezone you'd *guess* for each — then list, honestly, every alternative explanation (scheduler, shared account, deliberate misdirection) that would break your guess. The skill being trained is not the guess. It is the list of ways the guess is wrong.

Recap

- Resolve every handle to its **numeric ID** at the very start, and treat the ID as your subject from then on. The username is a mutable costume — now literally auctioned on Fragment — while the number is the person and never changes hands.
- Two reads come straight off the ID with no contact: account age (estimate, $\pm$ days–weeks — catches socks and coordinated batches) and name/username history (observation-bound — absence proves nothing).
- Find people through the **public groups** they post in: top-down via membership bots (undercount, verify by hand) or bottom-up by **intersecting rosters** you pulled yourself.
- **Activity timing** approximates timezone and routine — a slow accumulation of weak tells, confirmed on a second axis, never announced alone.
- **Lookup bots** are rate-limited, sometimes wrong, sometimes ToS- or breach-tripping. A caller-name can be an old owner. Every bot output is a lead to confirm.
- **The operator sees your query.** Every third-party bot hands your interest to whoever runs it — many are honeypots or adversary-run, and against an organized target this is how you get made. Prefer the official client and your own tooling; query only from a clean research account; treat convenience bots as burned by default.

- The durable pivot is often an old, un-scrubbed handle, not the clean current one — chase the persona the target forgot to bury.
- Work open surfaces only — no private groups under false pretences, no messaging the target, no logging links.



CHAPTER FOUR

Faces and Photographs

A photo is two clues in one — the frame and the face

A photograph is the most generous thing a person can hand you, and the most dangerous thing you can misread. Generous, because it carries two clues welded together. Dangerous, because the face pulls at something older than judgement — you *see* a match and your gut shouts *that's him* before your brain has done any work at all.

Two clues in one frame. First, everything *in* the picture: the metadata baked into the file, the shop sign over the shoulder, the reflection in the sunglasses. Second, the face itself — a key that

can, sometimes, open other doors across other platforms.

We treat both. The frame here is the *light* version — enough to read a photo for leads. Deep geolocation, the art of pinning a picture to a rooftop from a chimney and a shadow, is Volume 3. Here we stay at the level of "what does this photo tell me about who and where, and which of those threads can I pull."

And before any of it, one hard line. Face search is the most intrusive tool in this book. Nothing else you'll learn comes close.

A face template is sensitive personal data, and face-search tools sit in a legal grey zone that varies sharply by country. A client hiring you is not the law giving you permission. Check what's allowed where you work before you run one.

The frame: what's in the picture

Start with the file before you start with the face. Two reasons. It's less intrusive, and it often works better.

Every original photo can carry EXIF — a block of data the camera writes at capture. GPS coordinates. A timestamp. The phone's make and model. The editing app that last touched it. On an *original* file this is a gift.

The catch, and you must internalise this: social platforms strip it. Instagram, Facebook, X, WhatsApp — almost all of them wipe EXIF on upload. So does a screenshot. So does most

re-saving. If you pull a profile picture off Instagram and find no GPS, that tells you nothing about where the photo was taken. It tells you the photo went through Instagram.

```
EXIF METADATA READER
File: holiday_original.jpg
-----
Make       : Apple
Model      : iPhone 13
DateTime   : 2024-08-03 17:42:11
GPS        : 41.3874, 2.1686
Software   : -
-----
File: profile_pic_from_ig.jpg
-----
Make       : -
Model      : -
DateTime   : -
GPS        : (none)
Note: EXIF stripped on upload
```

Metadata reader — original vs platform-stripped

When GPS *is* present, treat the number as a claim, not a fact. GPS can be edited or spoofed. So you confirm it: drop the coordinate into satellite or street view and check the background matches — buildings, terrain, signage. If the metadata says Phoenix and the frame shows snowy peaks, believe the peaks, not the number.

The rest of the frame is read by eye and by reverse search. A badge with a name. A watermark username in the corner. A logo on a hoodie. A street sign. These "environmental anchors" are gold precisely because they often survive when the face fails —

and they're far less intrusive to chase.

Reverse image search, done properly

Amateurs upload the whole photo to one engine, get nothing, and quit. Here's the professional version.

Run several engines. They have different indexes and different algorithms. They are not interchangeable — a null result on one is not a null result. The rough division of labour:

- Yandex — the near-match workhorse. Best free tool for finding *other photos of the same person*, and strong on Eastern European and Cyrillic web. Its similarity is weighted toward facial geometry.
- Google Lens — the scene reader. Objects, landmarks, storefronts, text in the image. Weak on faces by design (it suppresses identity matches), strong on everything around the person.
- TinEye — the origin tracker. Finds exact and near-exact copies, and crucially can sort by *oldest*, surfacing where a picture first surfaced. No face recognition at all.
- Bing Visual Search — the extra pass. Different coverage, catches things the others miss. Commerce-heavy.

Try it (2026). Crop tight to the face, then run that crop through several engines that each do a different job (see the division of labour above). Then do the real work: open the *pages*

the hits live on and read the usernames, captions, and comments — never conclude on an engine's similarity score, which is a lead, not an identification.

Crop to the subject. This is the multiplier that turns a null into a hit. Most engines lock onto the biggest face or object in the frame. If your subject stands next to two other people and a large dog, the engine may cheerfully go and identify the dog. Crop tight to one face, one sign, one tattoo. Search each separately.

Search the face versus the whole image — deliberately, as two different questions. The whole image finds *this exact picture* elsewhere: reposts, the original source, catfish reuse. A tight face crop asks a different question: *where else does this person appear*, in other photos, other clothes, other years. You want both answers.

The flip trick. Scammers and repost accounts often mirror an image horizontally to dodge exact-match detection. Upload a flipped copy yourself as a second pass — new hits appear. Use flipping to *find reposts*, never to *compare a face*, because mirroring distorts the asymmetric features that matter.

A warning that follows from how these engines work: they return **similarity, never identity**. Every one of them will hand you doppelgängers without blinking. That's not a bug you can tune away.

The face as an entry point

Now the face itself. Given a face, dedicated face engines search for the same face elsewhere on the web.

Understand the mechanism, because it explains the failure modes. The engine detects a face, turns it into a numeric embedding — a vector of a few hundred numbers describing the geometry — and finds the nearest neighbours to that vector in its index. A "match" is a near neighbour in vector space. It is not a fingerprint. False-positive rates swing hard with pose, lighting, image quality, and demographic. The engine's own confidence score is not an identification.

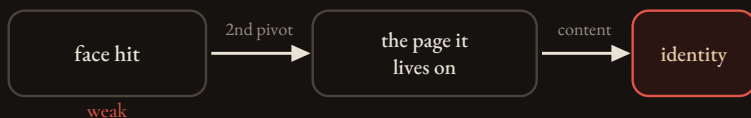
Coverage differs by engine, the same way it does for reverse search. Some index the open web — news, blogs, personal sites — and deliberately skip the big social platforms. Others lean into social and grey-area directories. Others specialise in regional networks. If your subject is Eastern European, a general engine may be thin where a regional one is rich.

Here is the rule that governs the whole discipline, and it is the throughline of this book:

A face hit is a lead. It stays a lead until content confirms it — never a rung higher on the strength of the match alone.

The identification doesn't come from the engine. It comes from the *second pivot* — moving from the face hit to the page it lives on, and reading it. The username. The captions. The comments. Who's tagged. A background you've seen in another photo of your subject. That's where a lead becomes a finding.

A FACE IS ONLY A LEAD



the engine finds a neighbour; the pivot makes the finding

A face hit only opens a door. The first pivot lands you on a page; the second pivot — reading the content on that page — is what actually identifies. The match score never climbs the ladder on its own.

One face, four doors, two of them wrong

Let me walk a case. Fictional throughout, but the shape is real — including the parts where it goes wrong, which is most of them.

A client engages me to vet a business contact before a deal. I have written authorisation, but the client's mandate is not the subject's consent — so before I run anything I confirm the work is defensible where I operate, the contact is a public-facing

principal named in the deal's paperwork rather than a private individual surveilled for sport, and I lean on open sources only. I have one thing to work from: a profile photo from a company "About" page. A man, call him Marco Feldt, mid-forties, plain headshot.

Step one, before any face engine: is this even a real person? Sockpuppets and scam profiles increasingly use AI-generated faces. So I look for the classic tells — asymmetric glasses, a warped hair boundary, garbled text in the background, liquid-looking reflections in the eyes. Marco's photo shows none of them. But in 2026 that settles nothing. The current generation of image models fixed those giveaways, and synthetic-image detection by eye is no longer reliable — the visible tells are mostly gone. So the honest reading is "no obvious tells, but that is not proof of a real person." I log it as an open question, not a cleared one, and I proceed with the real-versus-synthetic verdict still hanging over the case — something the later corroboration will have to settle, not my eyes.

I crop tight to the face and run it through two engines with different coverage.

```
FACE SEARCH — ENGINE A
Query: marco_feldt_crop.jpg
-----
Result 1  sim 94%  news-site
          caption: "M. Feldt, keynote"
Result 2  sim 91%  same site
Result 3  sim 88%  conf page
Result 4  sim 71%  stock photo
          -> flagged: possible reuse
-----
4 hits. 1 flagged low-confidence.
```

Open-web face engine — ranked hits

The top three cluster on one news site and a conference page, all naming M. Feldt at an industry event. That's coherent. The fourth, at 71%, points to a stock-photo library — I park it as suspicious and come back to it.

Engine two, with social coverage, gives me a fourth door:

```
FACE SEARCH — ENGINE B
Query: marco_feldt_crop.jpg
-----
Hit 1  score 82  X / handle
      @m_feldt_biz bio: "founder"
Hit 2  score 79  LinkedIn
      "Marco Feldt — CEO"
Hit 3  score 68  dating profile
      name: "Marc, 41" city: far
-----
Too few faces on Hit 3 to rank.
3 hits. 1 ambiguous.
```

Social-coverage face engine — ranked hits

Now I have three candidate accounts and two loose ends. Time to do the actual work — the pivots.

Door one, the doppelgänger. Hit 3 on engine B, "Marc, 41", on a dating profile in a different city. The face score is a plausible 68%. My gut, if I let it, would file this as "same guy, secret life." So I read the *content*. The dating profile's photos include a graduation shot dated years before Marco's known career, in a university Marco never mentions, in a country his verified LinkedIn history never touches. The dates don't line up. The place doesn't line up. I reject it. This is a doppelgänger — a real, different human who happens to sit near Marco in vector space. The face engine did exactly what it does: found a near neighbour. It was my job, not the engine's, to notice they can't be the same person.

*A high match score with a broken timeline isn't a secret.
It's a stranger.*

Door two, the dead end. Back to that 71% stock-photo hit. I open it. Marco's face, on a royalty-free image titled "confident businessman in office." Uh-oh — does this mean *Marco's* photo is fake, a stock image dressed up as a CEO? I run the flip trick and a couple more reverse-search passes on the stock image itself. TinEye, sorted oldest. The stock library's copy is old, but Marco's conference photos, from a named event with a dated programme, predate it — and show him live on a stage with other identifiable, verifiable people. What actually happened is dull: the stock library scraped or re-listed a real photo. The stock hit *links to nobody*. It's a repost with no new person behind it. A

reverse search that finds only reposts of a picture you already have has found you nothing. I close the door. No rung gained, no rung lost.

Doors three and four, the real ones. The X handle @m_feldt_biz and the LinkedIn "Marco Feldt — CEO". Here the pivots pay off. The LinkedIn lists the same company as the "About" page I started from. The X account, over months of posts, references the same conference from Marco's news hits — posts a photo *from* the event, the same stage, timestamped that week. Two independent threads — the professional page and the social account — now agree on employer, event, and timeline. I take the recovered handle and run it through a username tool to see where else m_feldt_biz lives, which turns up a matching profile on one more platform, self-authored, same bio.

Where does that leave me? The face alone was weak. Face plus one corroborating account would have been moderate. Face plus two independent accounts agreeing on employer, event, and dates, plus a consistent handle across platforms, is strong. Not confirmed — confirmed would want something harder, a direct statement, a document, the subject's own acknowledgement. I stop at strong and I write it down as strong, with the chain that got me there. I do not let the vivid 94% top hit tempt me into calling it confirmed.

An exercise you can run safely

Do not practise this on a stranger. The whole point of the chapter is that you don't. Practise on a *safe* subject, one of these three:

1. Yourself. Crop a photo of your own face and run it through a couple of reverse-image and face engines. See what the open web holds on you. Uncomfortable, instructive, and entirely consented. 2. A willing friend who says yes, in words, first. 3. A public figure with a genuine reason — a politician, a chief executive whose public role you're legitimately checking.

The task: take one photo. Read its frame first — is there EXIF, or was it stripped? Any anchors in the background? Then run it through at least two different reverse-search engines and, for a consenting or public subject, a face engine. For every hit, force yourself to write one line: *is this a real new signal, a repost of the same image, or a doppelgänger?* Chase one recovered username to a second account. Then place your conclusion on the ladder — weak, moderate, strong — and write the chain that justifies the rung.

If your honest answer is "weak," stop there and say so. Learning to stop at weak is the skill.

Recap

- A photo is **two clues**: the frame (metadata, background, anchors) and the face (a cross-platform key). Read the frame first — it's often more productive.
- EXIF is a gift on original files and **usually stripped** by social platforms. Absent GPS proves nothing. Present GPS is a claim to verify against the background, not a fact.
- **Reverse search properly**: several engines, crop to the subject, search the face and the whole image as *two different questions*, use the flip trick for reposts. Every engine returns *similarity, never identity*.
- A face hit is a lead, downgraded until content confirms it. The identification comes from the second pivot — the page, not the engine.
- **Expect dead ends**: doppelgängers rejected by broken timelines, stock or AI avatars that link to nobody, reverse searches that find only reposts. Rejecting these cleanly is the job.
- Land on a **defensible rung** and write the chain. Strong is a fine, honest place to stop. Confirmed needs more than a face.
- And the line that governs all of it: **face search is the most intrusive tool you own**. Written authorisation, a legal check, open sources only, and stop at the line.



CHAPTER FIVE

Leaks, Handled Cleanly

Breach data as a bridge between identities — never a key

A leak is not a shortcut. That is the first thing to unlearn.

Most people, when they first hear that a target's email and password are floating around in some dump, feel a little thrill of access. *I could get in.* Put that down. You never will, not as part of this work. What a leak gives you is far more useful and far less exciting: it gives you a *thread*. The same email in two breaches. The same rare password on two accounts. A username that shows up beside an address you already had. Threads that tie one identity to another. You follow them. You never pull them to

open a door.

You may check whether a door was left open. You may never walk through it.

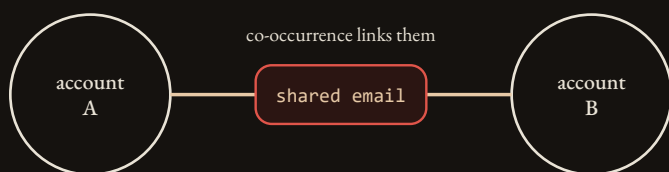
You already know that zero breach hits does not mean safe, only unindexed. Here we go deeper: what a breach checker actually tells you, how an email becomes a cross-platform connector, why a shared password can *fingerprint* a person, and the hard limits on every one of those moves.

The idea: leaks connect, they never enter

Think about what a breach dataset really is. A service got compromised, and its user records spilled — emails, sometimes usernames, sometimes passwords or password hashes, sometimes phone numbers and addresses. A *stealer log* is different again: it is the output of malware on someone's own machine, recording every site they logged into while infected. Two different things, and you must keep them separate in your notes, because they mean different things about the person. The 2026 shift worth naming: classic combolists are increasingly being crowded out by fresh stealer logs — Redline, Lumma, MetaStealer — traded on Telegram channels that parse them on the fly. The password-fingerprint method below does not change because of this; it still links accounts the same way, only the *source* has shifted.

But for attribution — for the work of proving that two accounts belong to one human — the content matters less than the *co-occurrence*. When `dmitri.works@gmail.com` shows up in a 2019 forum breach alongside the username `nightferry`, you now have a link between an email and a handle. When that same `nightferry` appears on a games platform and a photo-sharing site, the email you started with has quietly become a map of a person's footprint.

A LEAK IS A BRIDGE, NOT A KEY



you connect the dots — you never unlock the door

An email, a handle, or a password is a bridge between two identities — it lets you cross from one record to another. It is never a key: it does not, and must not, open the account at the far end.

None of this requires a password. And when a password *does* enter the picture, it enters as a *fingerprint*, never as a credential. I will come back to that, loudly.

Method: what a breach checker gives you, and what it hides

Start with the plainest tool. A breach-membership checker — Have I Been Pwned is the reference — takes an email and tells you which known breaches it appears in, when they happened, and what *classes* of data each one exposed. Passwords, dates of birth, addresses, phone numbers: it names the categories, not the values. HIBP will never hand you a password. That is by design, and the design is what keeps it lawful.

```
Account: dmitri.works@gmail.com
```

```
Breaches found: 3
```

- Collection #1 (2019-01)
Exposed: Emails, Passwords
- ForumRunner (2016-08)
Exposed: Emails, Usernames, IP addresses
- MyFitDiary (2018-03)
Exposed: Emails, Passwords, Names, DOB

What a breach-membership check returns

Read that carefully, because it is already telling you a lot without a single password. Three breaches across three years. One of them, ForumRunner, exposed a *username* — that is a pivot. Another exposed a *name and date of birth* — those corroborate an identity you may already suspect. The email is registered on at least three services, one of them a fitness diary, which places the person on that platform in 2018.

Now the part beginners skip. Run the same check on a slightly different address and you may get this.

```
Account: d.works1990@outlook.com  
Breaches found: 0  
No pastes found.
```

What a breach-membership check returns (empty)

Zero. What does zero mean? It does not mean the account is clean, safe, or fake. It means *the checker has not indexed a breach containing it*. The address may be pristine. It may be sitting in a dump nobody has parsed and loaded yet. It may be brand new. A blank result is a fact about the *index*, not about the *person* — data, not a verdict. Write it down as "unindexed," never as "no exposure."

There is one more thing these tools surface: *pastes*. A paste is a chunk of text someone dropped on a paste site — Pastebin and its cousins — often a raw list of leaked credentials or a combolist. When a breach checker says your email appeared in a paste, it is telling you the address was circulating in some dump that got copied into the open. Useful as a circulation signal. But pastes are messy, frequently duplicated, and often mislabelled with the wrong "source." Treat a paste hit as *moderate* evidence of exposure at best, and go read the paste's date and context before you lean on it. One 2026 caveat: paste ingestion has been largely frozen since paste sites tightened access, so this feed is stale and thin now — a blank "no pastes" line reflects a checker

that mostly stopped listening, not a clean subject.

Try it (2026). For a breach-membership check, go to Have I Been Pwned, enter an address you own, and read the returned breach list with its dates and exposed data classes. For password rarity, use the same site's Pwned Passwords range check, which returns how often a string has appeared without ever seeing your full password or hash. Services and interfaces shift, so confirm the current tool before you rely on it.

Method: the email as a cross-platform connector

Here is where a single email earns its keep. You have `dmitri.works@gmail.com`. The breach check gave you the username `nightferry` from ForumRunner. Now you fan out.

You take `nightferry` and you look for it across platforms — the username-enumeration tooling you met in Volume 1. You find a `nightferry` on a games service and a `night.ferry` on a photo site. Are they the same person? Not yet. A shared handle is *weak* evidence on its own; handles get copied, squatted, and reused by strangers. You need corroboration.

So you go back to the leak layer. The photo site's `night.ferry` — does that account tie to any email in a breach dataset? If a searchable breach index shows `night.ferry` was registered with `dmitri.works@gmail.com`, you have just closed the loop. The handle *and* the email co-occur in an independent source. That is no longer a coincidence of names; that is two records, two

platforms, one email. Your rung climbs from weak to moderate.

Notice what you did *not* do. You did not log into the photo account. You did not test a password anywhere. You read exposure and co-occurrence, and let the facts stack up.

Method: the password as a fingerprint (and only that)

This is the technique people most often get wrong, so I am going to be careful.

Some datasets expose passwords — as plaintext, or as hashes. Suppose two accounts, on two unrelated services, both used the password `Tramline!7743xQ`. That is not a common password. It is long, it is odd, it has no dictionary shape. If the same rare string protects two accounts, the most likely explanation is that *the same person set both*. People reuse their weird passwords. That reuse is a fingerprint.

You can use that. If `dmitri.works@gmail.com` used `Tramline!7743xQ` in one breach, and an account under a totally different email, `zorin.market@proton.me`, used the identical string in another, you now have moderate-to-strong evidence that both emails belong to one person. The password did the linking.

It did not open anything. And this is the line:

*You may use a leaked password to link two identities.
You may never type one into a login. Doing so is
unauthorised access — a crime, not OSINT.*

Now the limits, because the fingerprint has three sharp ones.

First: common passwords fingerprint nothing. If two accounts share Password123 or liverpool or qwerty2020, that is not a link — millions of people used those. A password only fingerprints when it is *rare enough that coincidence is implausible*. You can sanity-check rarity with a k-anonymity password checker, which tells you how many times a password string has appeared in breaches without you ever sending the password — or even its full hash — anywhere: you send only a short *prefix* of its hash as a range query and finish the match on your own machine. A password seen four million times is worthless. A password seen twice is a strong fingerprint.

Second: breach data is old and dirty. A password from a 2016 dump tells you about 2016. People rotate passwords. Combolists recycle and re-attribute records, so the "same" credential can appear across a dozen dumps with the wrong service attached — and in 2026 the stealer-log and combolist markets are bigger and more heavily re-attributed than ever.

Third, and the subtlest — this one has burned careful analysts, so I am naming it outright. A shared rare password does not by itself prove a shared *owner*. It can equally mean a shared *victim*.

Call it the shared-victim trap.

Picture it. Tramline!7743xQ shows up protecting dmitri.works@gmail.com in a 2019 dump, and then again on a crypto-exchange account in a 2023 dump. The tempting read is "same person, two accounts." But a second story fits the exact same facts: an attacker harvested Dmitri's rare password from the 2019 combolist and credential-stuffed it — tried that leaked email-and-password pair against sites Dmitri never used, the exchange among them. If it landed, the exchange account was *compromised*. It belongs to whoever Dmitri's credential was stuffed against, or to the attacker who opened it — not to Dmitri. Either way, the second account is a *victim of Dmitri's leak*, not a second identity *of* Dmitri. The password links two records; it does not tell you whether they share an owner or share an attacker.

So you break the tie before you conclude, never after. Check the order of events: if the second account was created or breached *after* Dmitri's password was already circulating in combolists, stuffing is live on the table and single-owner is unproven. Then look for ownership signals a stolen password cannot fake — does the second account carry its own recovery email, its own consistent handle, its own IP that overlaps the first? One password and nothing else, on an account that surfaced after the leak went public, is not a single-owner finding. It is a lead with two live explanations, and you hold both until something

independent kills one.

One worked case: from an email to a second handle

Let me walk a real-shaped case, dead ends included.

The seed. A client engagement — authorised, documented — hands me one email: `info@meridian-imports.co`. The question is who runs a particular anonymous marketplace account, `merid_trade`, that the client believes is connected to this small import firm.

Step one — breach check the seed. I run `info@meridian-imports.co` through a membership checker.

```
Account: info@meridian-imports.co
```

```
Breaches found: 2
```

- B2BLeadDump (2020-05)
Exposed: Emails, Company names
- ForumRunner (2016-08)
Exposed: Emails, Usernames, IP addresses

What a breach-membership check returns

ForumRunner exposed a username. Promising. I pull the associated record and it gives me the handle `merid_trade` tied to `info@meridian-imports.co`. For a moment this looks finished — the marketplace handle, right there beside the company email.

The dead end. It is not finished, and here is why. `info@` is a *role mailbox*. It is not a person; it is a shared inbox that several staff at

Meridian Imports can read and send from. Anyone in that office could have registered `merid_trade` using the shared address. Linking `merid_trade` to `info@meridian-imports.co` does not attribute it to a *human* — it attributes it to a *company inbox*. That is a real finding, but it answers the wrong question. The client wants a person. A role mailbox maps to no one person, and I mark this branch dead.

Step two — find a personal address. I need a human email. The ForumRunner IP-address field and the breach's timestamps let me see that the `merid_trade` account was most active from one IP, and that same IP appears in the breach tied to a *second*, personal-looking email: `dana.rurik88@gmail.com`. Now I have a candidate human.

Step three — corroborate with a second source. I breach-check `dana.rurik88@gmail.com`. It appears in a paste and in one other breach that exposed a password *hash*. Separately, I find a `dana.rurik` on a photo platform. Weak so far — a name overlap and an IP. I need something firmer.

Step four — the fingerprint, and a rejected coincidence. The breach tied to `dana.rurik88@gmail.com` exposed a password. So did an old account I found under the marketplace handle's recovery email. Both passwords: `summer2019`. My pulse jumps — then I stop. I run `summer2019` through the k-anonymity checker. It has appeared in breaches over sixty thousand times. Its fingerprints *nothing*. Half the internet used `summer2019`. I

reject the match outright. It is a coincidence, not a link, and if I had reported it as evidence I would have been wrong and reckless.

Back to slower, better proof. I keep reading. A different breach exposes, for `dana.rurik88@gmail.com`, the password `Rurik_kite#4402`. And the marketplace handle's associated email, in yet another dump, used the identical string. *That* password, checked for rarity, appears exactly once besides these two accounts. That is a real fingerprint — and before I lean on it, I clear the shared-victim trap: both accounts predate the dumps that leaked the string, both carry their own overlapping IP, and neither surfaced *after* the other's password went public. This is reuse by one hand, not stuffing by an attacker.

Step five — the hard identifier. Strong is not where this one ends, because a strand outside the leak layer finishes it. The photo-platform account, `dana.rurik`, is self-authored and public. Its bio links out to a small personal blog, and on that blog — in the author's own words, under the full name Dana Rurik — sits a post announcing the launch of the `merid_trade` marketplace shop, with a link straight to it. But a blog is a *self-set* surface: anyone can stand up a page under any name, so before I credit it I verify the blog's origin is independently hers. It is: the blog is linked from the `dana.rurik` photo account I already tied to her by the IP and the rare-password fingerprint, and its domain registration and inbound links trace back to that same account

— so this is a genuine self-link, not a namesake's page or one planted to mislead me. Only now does it count. That is not breach co-occurrence. That is the subject, in public, tying her own name to the very account I was asked about. A self-authored public statement, once its origin is confirmed as the target's own, is a hard identifier.

The landing. So where am I, honestly? The leak layer carried me a long way: an IP linking the marketplace handle to a personal email; a rare shared password, cleared of the shared-victim trap, linking that email to the handle's recovery address; a name overlap on a photo platform. Independent, stacked, none of them a login — and on breach data *alone* I would have stopped at **strong**, because stale, criminal-sourced records never earn *confirmed* by themselves. What lifts this to **confirmed** is the self-link that sits outside the leaks entirely: Dana Rurik, in public, under her own name, claiming the account. I write the report at confirmed, I name the hard identifier that earned the rung, I list every dump and date that got me to strong, and I flag the info@ dead end so the next analyst does not repeat my afternoon.

Exercise: check your own exposure, safely

This one you run entirely on yourself. No targets, no third parties.

- Take one of *your own* email addresses. Run it through a breach-membership checker (HIBP or equivalent). Write down each breach, its date, and the data classes exposed. Notice how much a stranger could infer just from the *list* — which services you used, roughly when.
- Now run a second address you rarely use. If it comes back empty, write "unindexed," not "safe." Sit with the difference. That blank is the exact trap you must never fall into when it is a target's address instead of yours.
- Take a password you have retired (never a current one) and run it through a k-anonymity password checker. See how many times it has appeared. If the count is high, that password could never have fingerprinted anyone — understand *why* in your own words.
- Finally, write one sentence: the difference between checking that a password was exposed and using it to log in. If you can say it plainly, you have the chapter.

Do not run any of this against another person's account, and never type a leaked password into a login form — not even your own old one on a live site to "see if it still works." Check exposure. Never test access.

Recap

- Leaks are bridges, not keys. They connect identities; they never open accounts. The bright line is looking versus touching, and it does not move.

- A breach checker tells you membership, dates, and data classes — never the password values. It reads exposure. That is what keeps it lawful.
- A blank result is data, not a verdict. Zero hits means *unindexed*, not *safe*, not *clean*.
- An email is a cross-platform connector. Breach co-occurrence turns one address into usernames, IPs, and alternate emails you can fan out and corroborate.
- A rare shared password fingerprints a person. A common one fingerprints no one. Check rarity before you trust it, and reject coincidences without mercy.
- A shared password can mean a shared victim, not a shared owner. Credential-stuffing can plant one person's leaked password on another's account. Clear the shared-victim trap — order of events, independent ownership signals — before you conclude single-owner.
- Role mailboxes name organisations, not people. *info@* and its kin map to no single human — never let one masquerade as attribution.
- Breach data is stale and dirty, and never earns confirmed on its own. Corroborate across sources, name every dump and date. Leaks can bridge you to *strong*; only a hard identifier outside the leak layer — a public self-link, a document — earns *confirmed*.



CHAPTER SIX

The Identity Graph

How to find someone who has hidden themselves

Here is a problem you will meet again and again.

You have a target. A name, maybe a photo, maybe a job title. You go looking, and you find nothing. Their Instagram is private. Their Facebook is a grey silhouette and a wall. Their LinkedIn is three lines. Every door is shut.

Most people stop here. They write "no online presence" and move on.

That is a mistake. Because the person did not disappear. They only locked their own front door.

One chapter back, you used a ring of connections to *confirm* an identity you already suspected — you had a name in mind and the network told you whether it fit. This chapter runs the other direction. Here you start with a person who has deliberately hidden themselves, and you *discover* an identity you did not know existed. Same graph of people; opposite job.

The move I reach for when the direct search dies is one I call the **Family Chain**. The idea is simple, and once you see it you cannot unsee it. You do not attack the wall. You walk around it, through the people standing beside it.

Never run the Family Chain to find someone who has gone dark for their own safety.

A domestic-abuse survivor or anyone hiding from the very person who wants them found is not a brief to fulfil — that "client" is a red flag to decline before you draw the first node. The full version of this limit is at the end of the chapter; it comes first here because a stop-sign is no use if you only meet it after the method.

The one thing they forget to hide

A careful person can lock down everything they own. Every profile, every post, every setting. It takes effort, but it can be done.

What they cannot do is lock down everyone connected to them.

Think about who surrounds a person. A wife. A husband. A sister. A mother. A brother. Old school friends. These people have their own accounts, their own habits, their own idea of privacy — which is usually *none at all*. They post the family holiday. They tag everyone by full name. They upload the birthday dinner. They leave an old phone number in a classified ad selling a sofa.

Your locked-down target appears in all of it. Tagged, photographed, mentioned, named. They did not consent to it and they may not even know it exists.

Be honest about how this has changed. By 2026 the platforms default to tighter privacy than they used to: tagging often needs approval, friend lists hide themselves, "only me" is a common setting, and a younger, more careful generation locks down by habit. The Family Chain is not the free-for-all it was a decade ago. But it still works, because it does not need *everyone* to be careless — it needs *one* relative to be. An aunt who never changed a setting since 2013. A cousin who leaves tagging open because approving tags is a chore. You are not betting on a

whole family being sloppy. You are betting on one open door in a house full of locked ones, and that bet still pays.

I once worked a case where a client was owed a large sum by a man who claimed his business had folded and he had nothing left. The man was a professional's professional online. Private everywhere. A spotless, sealed profile. Direct search gave me nothing. A normal analyst would have filed "no online presence" and closed the ticket.

I did not go at him. I went at the people around him. His wife's old page, half-forgotten and still open. From there, her public Instagram. In her Instagram, mentions of her sister. The sister's profile was wide open — and in a family holiday photo, she had tagged him. Not under his real, locked name. Under a *second* handle he thought no one knew about. That second account was where his real life — and his supposedly dead business — quietly carried on.

The method, not the story

I just told you a story, so distrust it a little. A story hands you one lucky ending and hides the ten steps that could have gone wrong. Let me strip it down to the repeatable method underneath: five steps you can run on anyone, without needing my luck.

Step 1 — Map the immediate circle. Before you pivot anywhere, draw the people. The target is the centre node.

Around them go the nodes you can name or guess: partner, parents, siblings, children, close friends, business partners. You do not need proof yet. You need candidates. A name from a wedding announcement, a "brother" in an old tagged comment, a shared surname — all of it goes on the map as a node.

Step 2 — Find which nodes are open. Now test each node for a pulse. Which of these people has a public profile? Relatives usually do. Older relatives especially — a mother's Facebook from 2013, wide open, is a goldmine. You are not reading them yet. You are just marking each node open or closed.

Step 3 — Pivot into the target through an open node. This is the move. You go into the open relative's account and you look *for the target inside it* — their tags, their photos, their mentions, the comments they left. You are not there to study the relative. You are there to find where they point back at your locked-down person.

Step 4 — Confirm by cross-links. One mention is a lead, not a fact. Stack independent links: same face across two photos, same event on the same date, the same handle in two unrelated places. Each match lifts you a rung.

Step 5 — Stop when the work question is answered. The moment you have what the job actually needed, you stop. You do not keep pulling the relatives apart for sport. More on this below, because it matters more than the technique.

Try it (2026). Pick a public figure with a locked-down personal presence — or, better, a consenting friend who has agreed to be your practice target. Write down three or four named relatives you can identify from public sources: a spouse, a sibling, a parent, an in-law, a cousin. Check each one's account for a pulse — mark them open or closed — and pick the most open relative. Now step *into* that relative's account and look only *for the target inside it*: tags, family photos, mentions, comments the target left. The moment you have your answer to the one work question you set, stop — do not record anything about the relative that isn't the answer, and do not read further "just to see." Caveat: this whole drill lives inside the safety carve-out at the end of the chapter — never run it against someone hiding for their own protection, no matter how neat the practice case looks.

Drawing the graph

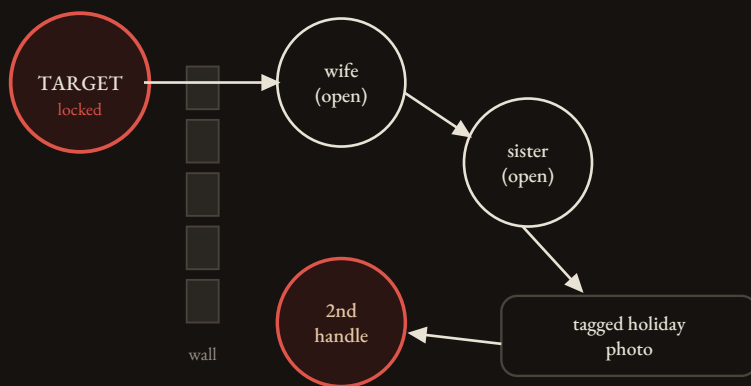
Put it on paper. A crude diagram beats a clever memory.

identity-graph.txt

```
TARGET (locked)
|
[ WALL ] <- direct search dies here
|
+--> WIFE ..... OPEN
    |
    +--> SISTER ..... OPEN
        |
        +--> family photo
                tags TARGET as
                a 2nd handle -> EVERYTHING
```

Read it top to bottom. The target is sealed. The wall stops the frontal search. So you step sideways to the wife, who is open, and sideways again to the sister, who is open — and the sister is where the target leaks. The arrows never go *through* the wall. They go around it.

THE FAMILY CHAIN



you don't climb the wall — you walk around it

The locked target sits behind a wall the frontal search cannot cross. The relatives stand out in the open around it. The path does not go through the wall — it runs sideways, from an open wife to an open sister to a tagged photo, and reaches the target from behind.

A worked case, with the wrong turns left in

Let me run it properly. Everything here is invented, but the friction is real — the dead ends, the look-alikes, the false link I had to throw away. Those are the parts most write-ups delete. They are the parts that teach.

The brief. A client is owed a large, court-recognised debt by a man named Marat Devlin. Marat swears his small logistics firm collapsed, that he is broke, and that there is nothing to recover. The client's lawyer suspects otherwise — that Marat quietly kept trading under a different name to put the business beyond reach. My question — the *only* question I am paid to answer — is: is Marat still operating a hidden business identity? There is an active, lawful dispute and a real debt behind the question — a genuine reason to look into Marat. Hold on to that, because the chain is about to run through his family, who are owed no such scrutiny.

Direct search. Dead on arrival. One LinkedIn, three lines, no photo. A Facebook with a grey head and nothing public. An Instagram set to private, 0 posts visible, 411 followers. He has done the work. The front door is bolted.

So I stop pushing on the door.

Step 1 — Map the circle. I scrape what little is public. His LinkedIn lists the firm; the firm's dormant "About" page names a co-founder, Devlin, same surname — likely a brother or a wife's byline. His locked Facebook, before it was locked, was cached once by a search engine with a single visible line: "married to Lena." So my map:

```
circle-map.txt
```

```
TARGET: Marat Devlin (locked)
- Lena Devlin ..... wife? node
- "Devlin" co-fndr .. sibling? node
- unknown parents ... node
- unknown siblings .. node
```

Four nodes. Two named, two blank. Good enough to start.

Step 2 — Who is open. I search each. "Lena Devlin" returns a public Instagram — travel photos, a dog, no privacy at all. The co-founder "Devlin" turns out to be Sam Devlin, and here is my first dead end: I assumed Sam was Marat's brother. Wrong. Sam's own posts make it plain he is a business partner, no blood relation, and he lives on another continent. He is a node, but he is not a *family* node, and family is where the leaks are. I mark Sam low-priority and move on. A guessed relationship that fails is not wasted time — it is one fewer wrong road later.

Step 3 — Pivot through the open node. Lena's Instagram is the way in. I do not sit and read her life. I look for Marat. And in her

feed, across two years, she mentions "my sister Kira" three times. Kira is a new node — a relative I did not have at step one, handed to me by an open account. This is the chain extending itself.

Now the second dead end, and a false link I nearly bought. I search "Kira Devlin" and find a fully open profile — public, chatty, lots of family photos. Jackpot, I think. Then I slow down and do the arithmetic instead of trusting the thrill.

First, base rate. How surprising is a matching "Kira Devlin"? Not very. Run the name through a public directory for the country and you find dozens of them — it is an uncommon-but-not-rare name, and "a Kira Devlin exists on Instagram" is close to guaranteed. A match this cheap is not evidence; it is what you'd expect by chance. To be worth anything, a candidate has to beat the base rate — she has to carry facts that a *random* Kira Devlin would almost never have.

So I check the independent facts, and two of them break the match outright. One: a sister shares a *maiden* name, and Lena's maiden name is not Devlin at all — so the real sister would not be a "Devlin" in the first place. The surname that pulled me in is the wrong surname to expect. Two: this profile's tagged family, home town, and school all belong to a different cluster of people entirely, none of whom overlap with Lena's. She is a namesake — a common name colliding by chance, nothing more. A shared name is only as strong as the name is rare, and this one is not rare

at all.

I reject her. Cheap match, contradicted by the details that matter, off the ladder entirely. If I had run with her I would have built a whole false identity for Marat out of a stranger's relatives.

Back to Lena. Her captions eventually give the sister's real handle — @kira.m.travels — no surname Devlin at all. That profile is open too. And in a beach photo from a family holiday, three adults are tagged. One tag reads @md_offgrid — a handle I have never seen on anything connected to "Marat Devlin."

Step 4 — Confirm by cross-links. One tag is a *weak* lead. I do not celebrate. I stack — and I am strict about what each piece is actually worth:

- The face in the beach photo (@md_offgrid) *looks like* the one blurry, low-resolution thumbnail cached from Marat's old Facebook. I want it to be him. But a blurry face is a lead, not a fact — Chapter Four's rule holds here too, and I refuse to let my gut cash a fuzzy thumbnail as proof. I write it down as a weak signal, a reason to keep looking, and nothing more. On its own it does not move me off weak.
- @md_offgrid's few public posts show a van with a company logo. The logo is the same mark — same wordmark, same colours — as the logistics firm on Marat's old LinkedIn. This one I did not set and neither did he; it is a specific, checkable artefact from a different source than the face. That

lifts me to moderate.

- The holiday's location and date match a "van's out west, back next week" note posted on the firm's public page in the same week. A third independent thread — timing, from yet another source — landing on the same person. That takes me to strong: @md_offgrid is very probably Marat Devlin's hidden, still-active account.

Notice where the weight came from. Not the face — the face never rose above weak, and I never let it. The climb from weak to strong is carried entirely by the two signals I could *not* have talked myself into: an independent logo and an independent date.

What the second account answered. @md_offgrid is not off-grid at all. The van is working — it is posting jobs, pickups, and a booking contact, all under a new brand, all in the weeks Marat swore the business was dead. That directly contradicts "the firm collapsed, there is nothing to recover." And *that is the answer to the brief*. The lawyer asked whether Marat was still trading under a hidden identity. On the open evidence, he is.

Now the honest part about the rung. I reached strong — three independent threads agreeing — but I did *not* reach confirmed, and I say so in plain words. Confirmed would mean a piece of proof that ties the human being Marat Devlin to that account beyond a competing explanation: a subpoenaed platform record, a payment trail, an admission. I cannot get those from

the open web, and I will not inflate strong into confirmed to please a client. So my report says exactly this: *strong evidence that @md_offgrid is Marat's active account; recommend the lawyer compel the platform and payment records to close the final gap.* The open-source work takes it as far as open-source work honestly goes, and then hands off to the people who can compel the rest.

I write the finding. Then I stop.

Where you stop, and why it is the real skill

The technique runs on innocent people's carelessness — a wife, a sister, who did nothing wrong and never agreed to be in anyone's investigation. That is exactly why it needs a hard limit, and one limit comes before all the others:

Never run the Family Chain to find someone who has gone dark for their own safety.

A domestic-abuse survivor, an estranged ex, a person who cut contact and does not want to be found — locating them can put them in real, physical danger, sometimes from the very person paying you. A "client" who wants you to find someone who deliberately hid from *them* is a red flag to decline, not a brief to fulfil. This is the one job you say no to, no matter the fee.

After that, the everyday discipline is one line: **take only what answers the work question, then stop.** I needed to know

whether Marat was still trading. Lena's travel history, Kira's relationships, the family's address — none of it answers the brief, so none of it goes in the report. I saw the rest; I do not use it or store it. And keep to open sources only: the whole method rests on what people left public, so you never befriend, deceive, or blag your way into a closed profile.

One legal line worth holding on to: your client's permission covers looking into their target, not cataloguing the private lives of the relatives you pass through. When the stakes are regulated — employment, credit, insurance, anything cross-border — that stops being something you eyeball and becomes something you get in writing, with counsel. The full playbook lives in Volume 4.

The analyst who cannot stop is not more thorough. They are just nosier, and eventually they cross a line that gets the whole report thrown out — and their reputation with it.

Exercise

Take a public figure with a locked-down personal presence — a founder, a mid-level politician, anyone who clearly manages their own privacy well. Set yourself one narrow work question, such as: *which city do they actually live in?* Not their address. Their city.

Then run the five steps.

1. Map their circle from public sources — spouse, siblings, parents. Draw the graph. 2. Mark each node open or closed. 3. Pivot through one open relative and look only for the target inside it. 4. Try to answer your one question, and write down which rung you reach — weak, moderate, strong, confirmed — and *why*. 5. Deliberately practise the stop: list three things you saw about the relatives that you are choosing **not** to record.

Bonus, and this is the important half: force yourself to find one false link and reject it in writing. A namesake, a wrong-guessed relationship, a face that turns out to be someone else. If you did not hit a single dead end, you were not being honest with the evidence.

Recap

- A person can lock their own profiles. They cannot lock the people around them.
- The Family Chain: do not attack the wall — go around it, through open relatives' tags, photos, and mentions.
- Five steps: map the circle, mark open nodes, pivot in through an open relative, confirm by stacked cross-links, stop at the work question.
- Relatives are still the open door, even under 2026's tighter defaults — you need one careless relative, not a careless family.

- Confidence is built from *independent* links you could not have arranged — an independent logo and date carried this case. A blurry face stays weak; the gut does not get to promote it.
- A shared name is only worth as much as the name is rare. Beat the base rate with independent facts, or reject the match.
- Expect dead ends: namesakes, wrong-guessed relations, false faces. Rejecting them is the job, not a failure of it.
- Your client's permission covers looking into the target, not cataloguing the relatives you pass through. Take only what answers the question, then stop. Regulated and cross-border detail: Volume 4.
- Never run the chain to find someone hiding for their own safety — that "client" is a red flag to decline.



CHAPTER SEVEN

Making the Signals Agree

One clue is a guess. Many independent clues that agree are an answer.

Beginners hunt for the killer clue. The one screenshot, the one leaked email, the one detail that ends the argument. They think attribution is a lock and somewhere there is a single key.

It almost never works that way.

Real attribution is quieter. It is a pile of small, unremarkable observations — when someone posts, how they spell, which emoji they overuse, who keeps showing up in their replies — that on their own prove nothing, but *together* point one way and

refuse to point any other. No single signal carries the case. The agreement between them does.

A conclusion built on one clue falls over when that clue does. A conclusion built on five independent clues has to be knocked down five times.

Volume 1 gave you the ladder: weak, moderate, strong, confirmed. This chapter is about how you actually *climb* it — not by finding stronger clues, but by stacking honest ones and watching them agree. And it is about the one mistake that quietly wrecks more attributions than any other: counting the same signal twice.

One thing to settle before we start, because it trips people up. You will see those same four words — weak, moderate, strong, confirmed — used in *two* places, and they mean two related things. They grade an **individual signal**: how much weight this one clue carries on its own. And they grade your **overall conclusion**: how confident you are after stacking everything. The two are linked by counting — but count is a rule of thumb, not the test. The Bridge chapter set the rule for the conclusion. What earns *strong* is that the independent agreements are, together, hard to explain by any rival hypothesis. That usually takes three or more — but two heavy, diagnostic signals (a byte-identical avatar plus a verified self-link) can reach it, and three weak, common ones may not. Two agreeing independent

signals is the usual floor for *moderate*. So a single "strong, independent" signal is not the same as a strong *conclusion* — it is one heavy clue that, with one or two more, gets you to a strong finding. When I label a signal below, I am rating that clue; when I name a rung at the end of a case, I am rating the pile. Keep the two senses apart and the chapter stays clear.

The idea: independent agreement is what moves the needle

Two accounts. You want to know if the same person runs both. You will never get a certificate. What you get instead is a set of signals, each of which is a little bit consistent — or a little bit inconsistent — with "same person."

The magic word is *independent*.

Two signals are independent when they come from *different sources* and one could be true while the other is false. When they agree anyway, that agreement is hard to explain by chance, and your confidence climbs fast. When they are *not* independent — when they secretly come from the same source — they feel like two witnesses but they are one witness talking twice. Stacking them fools you into false confidence.

Here is the cleanest way to feel the difference.

Say two accounts both display the name "Andrii Melnyk."

Impressive? No. The display name is *self-set*. The person typed it

into both boxes. If it is the same person, of course it matches — but if it is an impersonator, it also matches, on purpose. The display name agreeing tells you almost nothing, because both accounts' names come from the *same single act of typing by whoever controls them*. One source. One signal.

Now say account A posts almost only between 22:00 and 02:00 Kyiv time, and account B — on a different platform, which you checked separately — *also* goes quiet all day and lights up after 22:00. Neither account announced its hours. You measured them, independently, from two different timelines. Their agreement is not something a person set on purpose. That agreement is real signal.

Same-looking evidence. Completely different weight. The question is never "do they match?" It is "do they match *for reasons that could have come apart, and didn't?*"

Now the part most people miss, and it is the whole reason this chapter exists. "Did the target set it?" is only *half* of independence. There is a second, sneakier way signals stop being independent: when two clues share a hidden common cause that would make them agree *even for two different people*. Ask the real test out loud — would these two signals agree by coincidence for two different people drawn from the same population? If the answer is yes, they are not two independent votes. They are one vote wearing two coats.

Here is the trap in one picture. Suppose your true rival hypothesis is not "same person versus a random stranger" but "same person versus *a different person from the same city and the same online scene.*" Now think about what two such people would share *by default*: the same timezone, so the same posting hours. The same local slang. The same in-group emoji. The same regional spelling quirks. None of that is because they are the same person — it is because a shared city and a shared subculture is a common cause that pushes both accounts toward the same look. Under that hypothesis, timing and slang and emoji and typos are all downstream of one thing. Counting them as four independent hits is counting one coincidence four times.

The five workhorse signals

These are the signals you will reach for again and again. None is decisive alone. Learn what each is worth and, more importantly, whether it is independent of the others you already have.

- **Activity timing.** When someone posts, over weeks, sketches their timezone and daily routine — the night-owl window, the lunchtime lull, the dead zone while they sleep. It is behavioural and hard to fake, because faking it means never slipping for months. Two accounts with the same posting fingerprint across many days is quietly strong. *Independence: high against a random stranger — you measured it, they didn't set it. But careful: two different people in the same city*

share a timezone, so a broad "posts at night" agreement is weaker than it looks. What earns the weight is a specific, personal signature — the same odd 03:40 spikes, the same dead Sunday mornings — not just "both nocturnal."

- **Writing style (stylometry-lite).** Not fancy software — just the tics a person can't stop doing. A habitual "tbh," a comma before "but," British spellings mixed with American ones, a signature typo, starting sentences with "So." One tic is a coincidence. Five specific, unusual tics shared across accounts is a fingerprint. *Independence: moderate to high, if the tics are genuinely distinctive — but beware, people in the same subculture share slang.*

- **Emoji and punctuation habits.** A sub-species of style, worth its own line because it is so sticky. The exact emoji someone reaches for (vs), double spaces after full stops, "!!!" versus a lone "!", lowercase-everything. Muscle memory. People change their name before they change their emoji. *Independence: moderate — same caveat as style.*

- **Reused assets — bios, links, avatars.** The same profile photo, the same bio sentence, the same linked website or Telegram handle across accounts. This is high-value *but read the independence carefully.* A reused avatar file (byte-for-byte identical, not just similar) is strong. A reused *self-written bio line* is weaker than it looks — it is self-set, like the display name, so an impersonator can copy it deliberately.

- **Shared connections.** The same friends, repliers, group co-members orbiting both accounts. A person's social graph is expensive to fake and slow to move. When two accounts share not one but a *cluster* of the same real associates, that is one of the strongest signals there is — because it lives on *other people's* pages, outside the target's control. *Independence: high.*

Notice the pattern. The signals you *observe from the outside* — timing, graph — tend to be independent and strong. The signals the account *sets about itself* — display name, bio text — tend to be non-independent and weak, because whoever controls the account controls them directly, honestly or not.

So here is the rule that turns this from a vague worry into arithmetic — and it is the single most important rule in the chapter for the counting to come:

****Timing, slang, emoji, and typos are one family of signals, not four. Grade the whole family as a single factor, and set that factor's weight by how *rare* the shared trait is — not by how *many* traits you can list.****

So four agreeing tics from one subculture is not four rungs of evidence — it is one observation, "these two look like they come from the same milieu," counted once. The rescue is *specificity*. A widespread tic ("posts at night," "says based") proves little and cannot be laundered into strength by piling more widespread tics beside it. But an odd, personal one — a consistent unique misspelling, a homemade contraction nobody else uses, an emoji

nobody in that scene reaches for — is rare enough that a coincidental match across two strangers is genuinely unlikely. *That* is what carries weight, and one rare tic can be worth more than ten common ones.

So independence has two questions now, and you ask both of every signal:

1. Did the account owner set it? If yes (display name, bio text), it is self-set and weak — an impersonator copies it on purpose. 2. Would it agree by coincidence for two different people from the same population? If yes (shared timezone, in-group slang), it is correlated, and it collapses into one factor with its cousins.

A signal only counts as a full, independent vote when the answer to both is no.

The method: build a signal table, then test independence

Do not hold this in your head. Bias lives in your head. Put it in a table.

For each signal write four things, and keep the last two in *separate* columns because they are separate questions. One: what it is. Two: the value you observed on each account. Three: strength — how much this one clue is worth on its own (weak, moderate, strong). Four — the column everyone forgets — independent?, meaning both of our tests: not self-set, and not a

coincidence two same-population strangers would share. Strength and independence are not the same axis. A signal can be strong-looking yet non-independent (a copied bio), or modest yet perfectly independent (one odd typo). Never fold them into one column.

INDEPENDENT SIGNALS

signal	kind	independent?
display name	self-set	NO
bio text	self-set	NO
posting hours	observed	yes
social graph	observed	yes
slang + emoji	one style	count once

cross out self-set rows before you count — quantity is not independence

Two piles. On the left, signals the account sets about itself — display name, bio, avatar choice — which an impersonator copies on purpose. On the right, signals you observe from outside — posting rhythm, who replies, a rare typo — which the owner cannot quietly coordinate. The table's job is to sort every clue into the right pile before you count, and the "independent?" column is where that sorting is written down.

Here is a real one, from the case below.


```
signal-table.txt
```

```
kv_designs vs kateryna.v.art
```

signal	strength	indep?
display name	weak	NO self-set
bio text	weak	NO self-set
avatar (bytes)	moderate	YES observed
posting hours	moderate	YES *style-fam
emoji " " close	moderate	YES *style-fam
typo "wich"	strong	YES *style-fam
bio link	moderate	NO! see note
shared repliers	strong	YES 3rd-party

```
-----  
* style-fam: timing+emoji+typo are  
  correlated -> COUNT AS ONE factor,  
  weighted by the rarest (the typo).  
! bio link is self-set, BUT a replier  
  posted it first -> promote to YES.  
-----
```

*Strength and independence are separate columns: a signal can be strong yet non-independent, or modest yet independent. The three *style-fam rows collapse into one factor before you count.*

*Reading the marks: indep? asks "is this an independent vote?" — YES means it counts on its own, NO means it does not. *style-fam tags the one correlated style-family (timing, slang, emoji, typos) that gets counted once, not per row. 3rd-party means the signal lives on other people's pages, outside the target's control. The ! is a note flag — it points you to the numbered note under the box.*

Read that table the way I do, in two passes. First pass: cross out the self-set rows. Display name, bio text, and — at first — the bio link, because those are things the owner simply typed and an

impersonator could copy. Gone.

Second pass, and this is the new discipline: I do *not* count the three *style-fam rows as three votes. Posting hours, the closing emoji, and the typo are correlated — two different people from Kateryna's city and design scene could share the night-owl hours and even the plant emoji by coincidence. So I collapse the family into one factor and weight it by its rarest member: the misspelling "wich," which is odd enough that a chance match across two strangers is genuinely unlikely. That one factor lands at moderate-to-strong on its own.

So what actually survives to be counted? The identical avatar file (moderate, independent). The style-and-timing family, counted once (moderate-to-strong). The cluster of shared repliers living on other people's pages (strong, independent). Plus the promoted bio link. That is three-to-four genuinely independent agreeing factors — not the seven rows the table started with, and not the "five signals" a careless reading would claim.

The bio-link row has a star, and this is the subtle move in the other direction. On its own the link is self-set — worthless. But I found that one of the *shared repliers* had posted that same Linktree link months earlier, tagging Kateryna, before either account displayed it. That gives the link an independent origin outside the account, so I promoted it to YES. The discipline cuts both ways: you demote correlated tics that only look like many, and you promote a self-set field when its origin turns out to be

independent. You never grade a signal by how it looks — only by where it *came from* and whether it could have agreed by chance.

A worked case, with a seductive coincidence

A client ran a small design studio. A second account, *kateryna.v.art*, was posting their portfolio images as its own and taking commissions. The client swore it was a former freelancer, Kateryna, who had left on bad terms. My job: is *kateryna.v.art* actually Kateryna, or someone impersonating her to smear her — or just a random content thief? Fictional names, real method.

The seductive coincidence, first. Both accounts displayed "Kateryna V." and both bios read, word for word, "designer · plants · slow living." My client waved this as proof. It felt like proof. It was not — and this is exactly the trap the chapter is about.

The display name and the bio are *self-set*. If this were an impersonator building a fake Kateryna, they would copy the name and bio *on purpose* — that is the whole point of impersonation. So the match is equally consistent with "it's really her" and with "someone is pretending to be her." A signal that fits every hypothesis equally discriminates between none of them. I wrote it in the table, marked it **NO** — not independent, and refused to let it move my confidence a single rung. My client hated that. Good.

Then the real work. I pulled the avatar files from both accounts and compared them byte-for-byte. Not "look alike" — identical file, same asset. That is harder to explain by an impersonator, who would more likely screenshot or re-crop. *Moderate, independent.*

Then the family of style-and-timing signals, and here I am deliberately careful *not* to overcount. I logged posting times for three weeks: both accounts lived in the same 21:00–01:00 window and both went quiet on Sunday mornings. Both ended most posts with a single . And both misspelled "which" as "wich" — the same specific slip, twice from A, three times from B. Now, a lazy analyst writes that up as three separate strong hits. I do not, because they are correlated: a different designer from the same city and the same plant-loving corner of the internet could plausibly share the night hours and the emoji by coincidence. So I fold timing, emoji, and typo into one factor and weight it by its rarest, most personal member — the misspelling "wich," which no shared subculture explains. As one factor, it lands at *moderate to strong, independent*. One factor, not three.

The dead end. I got excited about a shared phone number. [kateryna.v.art](#) had a Telegram link, and the client gave me Kateryna's old number from a contract. They matched. For an hour I thought I was done.

Then I checked *where the number on the account came from*. It was pulled from the studio's own public "our team" page — which had listed Kateryna's number while she worked there, and still did. So the impersonator hypothesis explained it just as well: anyone could have lifted that number from a page the studio itself published. The number was not independent of the public record. It fit both stories. **Rejected** — struck through in my notes, "not diagnostic, publicly available." A clue that fits both of your hypotheses is worth exactly nothing, no matter how much it feels like a win.

What held. The clincher was the graph. I mapped the regular repliers on both accounts. Four of the six most frequent names were the *same real people* — a former classmate, two ex-colleagues, a sibling — and on the classmate's own public wall, she had replied to *kateryna.v.art* with "Kate, is this your new page?" and gotten a -signed "yeah, finally!" back. That exchange lived on *someone else's* page, outside either account's control, and it self-linked the account to the real person.

Stack it, counting honestly — factors, not rows, and no witness twice. Identical avatar file (moderate, independent). The style-and-timing family, counted once and carried by the rare typo (moderate to strong). The shared cluster of associates — but the classmate counts once. I credit her wall exchange, so I do not also count her inside the associate cluster; the cluster contributes the *three other* real people (the two ex-colleagues and

the sibling), which is still strong. And the classmate's public self-confirmation on her own wall (strong), a near-self-link the account owner could not have staged. That is four genuinely independent agreements — not the six or seven a row-by-row tally would have inflated it to, and not five, because the classmate would have been double-counted. Four independent factors all agreeing, with a near-confession among them, is well past the three-plus floor that earns strong. I reported confirmed — same person, this is Kateryna's own account, and I could defend every rung, including the ones I refused to double-count.

And I told the client plainly: the name and bio matching, the thing *you* found most convincing, was the one piece of evidence I threw out. It proved nothing. The boring stuff — when she posts, who answers her — proved everything.

Exercise

Take two accounts you already believe are the same person — a friend's Instagram and their Twitter, say. Now audit your own belief properly.

- Build the signal table with four columns: signal · observed value on each · strength · independent? Keep strength and independence in separate columns — they are different questions.
- For every row, ask both hard questions. Did the owner self-set it? And could two different people from the same city

or subculture share it by coincidence? A yes to either means NO in the independence column.

- Now group your correlated signals. Any timing, slang, emoji, and typo agreements go into a single style-and-timing factor, weighted by the *rarest* trait in it — not counted one by one.
- Cross out every NO row and collapse the correlated family. Count what is left. How many genuinely *independent* agreeing factors do you actually have?
- Find one signal you assumed was independent and prove it isn't — trace it to a single source (a self-set field, a public page both copied from, or one shared subculture).
- Write your honest rung using only the surviving independent factors. Is it lower than your gut said? It usually is.

The lesson lands when your confidence *drops* after the audit. That drop is the method working. It means you were about to count a witness twice.

Recap

- Attribution is not one killer clue; it is many small, independent signals that agree.
- Independence has two tests, not one. A signal is independent only if the owner did not self-set it *and* two different people from the same population would not share it

by coincidence. It must fail both to count as a full vote.

- Non-independent signals — anything the account *self-sets*, like display name or bio text — feel like evidence but discriminate between nothing. An impersonator copies them on purpose.
- Correlated tics are one factor, not many. Timing, slang, emoji, and typos move together for people from the same city or subculture — grade them once, by how *rare* the shared trait is, never by how many you can list.
- The four ladder words grade two things: an *individual signal's* strength and your *overall conclusion*. Two independent agreements usually earn moderate; three or more usually earn strong — but count is only a rule of thumb. What actually earns *strong* is that the agreements are, together, hard to explain by any rival hypothesis. A single "strong" signal is not a strong finding.
- Build a signal table with separate **strength** and **independent?** columns. Cross out the self-set rows, collapse the correlated family, then count what survives.
- A clue that fits *both* of your hypotheses equally is not diagnostic — reject it, however good it feels.
- The strongest signals live on *other people's* pages, outside the target's control. Weight them accordingly.



CHAPTER EIGHT

Fighting Your Own Certainty

The account you most need to investigate is the one running the investigation.

You have a good attribution. Several independent signals, all agreeing, a clean climb up the ladder. It feels finished. You can picture yourself writing "confirmed."

That feeling is the most dangerous moment in the whole job.

Because by now you *want* it to be true. You have spent hours on it. You told the client you were close. Your brain, helpfully, has stopped looking for problems and started defending the answer.

Every new detail somehow fits. Nothing surprises you anymore. That is not the case getting stronger — that is you going blind.

Volume 1 told you to watch your own bias. This chapter is the drill for actually doing it — a set of concrete moves that force you to attack your own conclusion before anyone else can. The goal is not to feel confident. The goal is to *deserve* your confidence, and to be able to say "cannot attribute" out loud when you don't.

The idea: try to break your own answer, on purpose

The instinct, once you have a theory, is to go collect more evidence *for* it. This feels like diligence. It is actually the trap. Every investigation has enough noise that a determined person can always find one more thing that "fits." Pile up enough confirming scraps and even a wrong theory looks armoured.

The fix is to invert the search. Instead of asking "what else supports this?", you ask the question that actually tests a theory:

What would I expect to see if I were wrong — and do I see it?

This is the disproof pass. You state your conclusion as a hypothesis, you imagine the world in which it is false, you list what that false world would leave lying around, and then you go looking *for that*. If you hunt hard for the disproof and genuinely

can't find it, your conclusion survives — and now it is battle-tested, not just comfortable. If you find the disproof, you just saved yourself from publishing a mistake.

It is the same logic as the competing-hypotheses discipline analysts use: you don't win by racking up support for your favourite, you win by *eliminating the alternatives*.

THE DISPROOF PASS



try to break your own answer — keep what survives

Two live hypotheses — "same person" (A) and "two people, one subculture" (B). A signal that fits both explains nothing; only a signal that fits one and breaks the other moves you. The disproof pass hunts for the fact that kills your favourite — and here, one does.

This matters more the longer you've worked a case. Effort quietly becomes ownership. Ten hours in, the theory isn't just a theory anymore — it's *your* theory, and knocking it down feels like admitting you wasted the ten hours. That is why the

disproof pass has to be a *procedure*, not a mood. Good intentions won't save you; you already have good intentions and you're still biased. What saves you is sitting down, writing the false-world checklist, and grinding through it whether you feel like it or not — especially when you don't.

The method: four moves before you conclude

Run these in order, every time the answer starts to feel certain. They take twenty minutes and they will save your reputation.

1. State the hypothesis in one flat sentence. Write it down, plainly. "Account B is run by the same person as account A." Naming it turns a warm feeling into a claim you can test. A claim you never wrote down is a claim you can't attack — and can't be held to.

2. Do the disproof pass — "what would prove me wrong?" List the observations you'd expect *if the theory is false*, then go look for each. If A and B are the same person, they should never post at the same minute from incompatible places. So: did they ever post *simultaneously*? Does B know something A couldn't? Did B keep going during a week A was verifiably offline? You are trying to catch your own theory in a contradiction. Hunt like you're being paid to break it.

3. Write the alternative explanations and take each seriously. "Same person" is rarely the only story. Force yourself to name the others, because a coincidence you never articulated is one

you can't rule out:

- Two different people who genuinely share a subculture, a slang, a home city — so their style and timing look alike for boring reasons.
- A shared account — a couple, a team, a small business — so the "person" is two hands on one login, and your stylometry is an average of both.
- A recycled handle — the username was abandoned and re-registered by a stranger, so "the account's history" is actually two people's histories stitched together at the handover date.
- A plant or sockpuppet — someone *built* B to look like A, on purpose, copying the self-set signals to frame or impersonate. (This is exactly why Chapter Seven made you throw out self-set fields.)

For each alternative, ask: what evidence would I expect it to leave, and have I actually ruled it out — or just not thought about it?

4. **Separate fact from inference in your notes.** Go through your file and mark each line as one or the other. *Fact:* "both accounts posted 21:00–01:00 across three weeks" — you observed it. *Inference:* "therefore they're in the same timezone" — you concluded it. Inferences can be wrong while the facts stay true (a scheduler, a VPN, insomnia in a different zone). When you blur

the two, a chain of guesses starts to read like a chain of evidence. Keep them in different columns so you always know which is load-bearing.

Here is what that separation looks like on paper.

FACT (observed)	INFERENCE (concluded)
-----	-----
both post 21-01	same timezone
both spell "wich"	same writer
B knew A's new phone	B is close to A
A silent Mar 3-9,	??? -> see disproof
B posted daily	
-----	-----
flag: row 4 is a CONTRADICTION, not a supporting fact. do not file it under "evidence for."	

case notes — keep the columns apart

fact and inference kept in separate columns

Look at that fourth row. It is a fact that does *not* fit the theory — A was verifiably offline for a week while B posted every day. A confirming mind skips past it. The disproof pass forces it into the light, in its own column, flagged. That row is where the real test of the case now lives.

Try it (2026). Take an attribution you currently believe — your own live case, or a public one you’ve seen confidently asserted — and write down two questions on a blank page: *"what would I expect to see if I were wrong?"* and *"what one fact would kill this?"* Then spend twenty minutes hunting only for that killer fact — not more support, not "colour," just the disproof. When the

twenty minutes are up, re-grade your rung honestly against what you found (or didn't). If your confidence dropped a rung, or landed on "cannot confirm," the pass worked exactly as designed. This is the single most professionally useful habit in the book — the one that, more than any tool, separates people whose findings survive challenge from people whose findings don't.

The calibration problem: weak piles are not strong

There is a slow, quiet failure mode that catches careful people. You gather ten weak leads. None is strong. But there are *ten* of them, and the sheer heap starts to feel like a strong case. So "weak × 10" silently becomes "confirmed" in your summary, and nobody — including you — notices the promotion happen.

Chapter Seven already told you why: ten leads that trace back to one source are one lead counted ten times.

So don't ask how many signals you have. Ask how many *independent* ones — and whether their agreement is hard to explain unless the accounts are one person. Write the honest rung. A defensible "strong" beats an indefensible "confirmed" every single time.

A worked case that does not resolve

A newsroom asked me to confirm that an anonymous account, @civic_watch_ua, posting sharp criticism of a local council, was really a specific council employee — call her O. — leaking from the inside. It *felt* certain within an hour. That feeling is why I'm telling this story. Fictional details, real method.

The seductive stack. @civic_watch_ua posted only 22:00–01:00 — O.'s known off-hours. It used specific procedural jargon a council insider would know. It once referenced a budget line before it was public. And its bio said "views my own · public servant." Timing, insider knowledge, a near-confession in the bio. The reporter was ready to run it.

I wrote the hypothesis flat: "*@civic_watch_ua is O.*" Then I ran the disproof pass, and it came apart in my hands.

Disproof, move by move. *What would I see if this is false?* First: the bio. "Public servant" is self-set — O. would write it, but so would anyone wanting the account to *read* as an insider, including a rival trying to get O. blamed. It fit "it's O." and "someone's framing O." equally. Not diagnostic. Struck out.

The budget-line "leak" felt strong until I hunted the alternative. I found the same figure in a committee circular that had gone to *forty* people. Insider knowledge, yes — insider to forty, not to one. It narrowed the author to a large room, not to O.

Downgraded to weak.

Then the disproof landed a real hit. I checked for the contradiction — did the account ever run while O. was verifiably elsewhere? O. had posted holiday photos on her own real, named account from abroad, phone-timezone visible, for nine days in one stretch. @civic_watch_ua posted through all nine, in the *same* 22:00–01:00 Kyiv window, never shifting to holiday hours. Two readings: either it isn't O., or O. scheduled posts, or it's a shared account and the *other* hand kept posting. I could not tell which. That single fact broke the clean story and I could not un-break it.

The alternatives, taken seriously. *Two people:* plausible — council slang and off-hours fit any of several insiders. *Shared account:* plausible — could be a small group of employees, which would explain posting straight through O.'s holiday. *Recycled handle:* checked, ruled out, the account was too young. *Plant:* genuinely possible — a rival faction had motive to make leaks look like they came from O. I could eliminate none of the first, second, or fourth.

The honest outcome. I had timing that fit O. but also fit three other stories. I had insider knowledge shared by forty people. I had a self-set bio worth nothing. And I had one hard fact — posting straight through her verified absence — that actively fought the theory. Separating fact from inference, my facts were real but every bridge to "it's O." was an inference with a living alternative underneath it.

I reported it as I found it: cannot confirm — do not conclude. Weak-to-moderate that the author is a council insider; *no* defensible attribution to O. specifically; a live, unexcluded possibility that the account was built to *implicate* her. I told the newsroom plainly that naming O. would not survive a challenge and could hand a smear the credibility of their masthead. They held the story. Months later a court filing named a three-person group — none of them O. My "cannot confirm" aged well. A "confirmed" would have been a disaster with my name on it.

That is the chapter's whole point. The disciplined answer is sometimes "I don't know," and delivering that answer cleanly — refusing to let a warm feeling and a thick file bully you into a rung you can't defend — is not failure. It is the most professional thing you will ever do.

"I cannot attribute this" is a finding. Say it with the same confidence you'd say "confirmed."

Exercise

Take an attribution you currently believe — one that feels solid. Now try to destroy it.

- Write the hypothesis as one flat sentence. No hedging, no feeling — a claim.
- Run the disproof pass: list what you'd see if it were false, and go hunt each item. Did the accounts ever post simultaneously? Does one know something the other

couldn't? Any contradiction you've been skating past?

- Write down all four alternatives — two people, shared account, recycled handle, plant. For each, name what would rule it out and whether you actually have that.
- Split your notes into two columns: fact you *observed* versus inference you *concluded*. Count how many of your bridges to the conclusion are inferences with a living alternative underneath.
- Recount your *independent* signals — not observations, independent ones. Now write the rung. If it dropped, or if it landed on "cannot confirm," you did the exercise right.

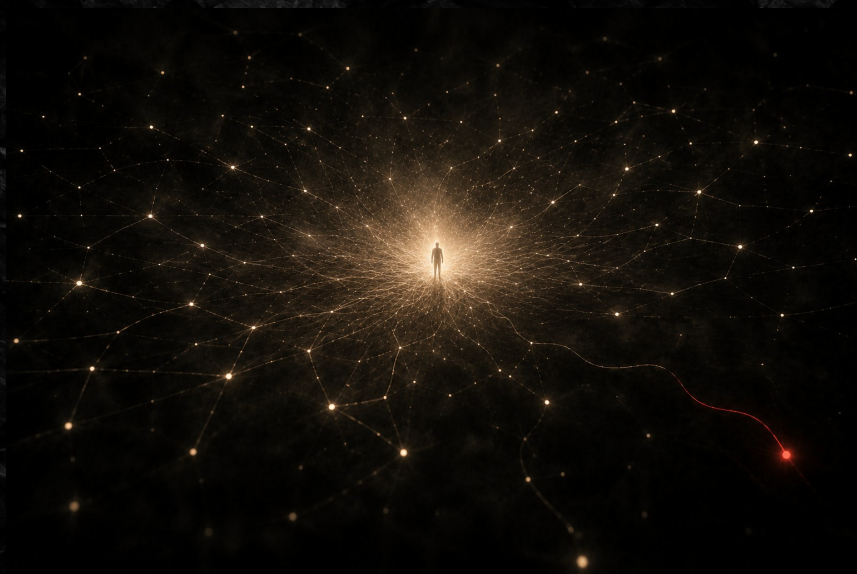
If your certainty survived a genuine attempt to kill it, it was real. If it didn't, you just dodged the mistake that ends careers.

Recap

- The biggest threat to an attribution is the analyst wanting to be right; certainty is a warning light, not a finish line.
- Invert the search: don't gather more support, ask "*what would prove me wrong?*" and go hunt the disproof.
- Four moves before concluding: state the hypothesis flatly, run the disproof pass, name and test every alternative (two people / shared account / recycled handle / plant), and separate fact from inference in your notes.
- A pile of weak or non-independent leads is not a strong case. Quantity isn't independence; a hundred rumours from

one mouth is one mouth.

- "Cannot attribute" is a legitimate, professional finding. Deliver it cleanly rather than forcing a conclusion the evidence won't hold.



CHAPTER NINE

The Full Attribution

Every thread, woven at once

Everything so far has been a single thread pulled slowly. A handle. A face. A leaked email. One technique at a time, in clean light.

Real work is not like that. Real work is eight threads at once, most of them fraying, two of them lying to you, and a clock. This chapter is one case, start to finish, the way it actually happens. Messy on purpose.

The question was small. The answer took a week.

I will show you the dead ends too. A case with no dead ends was not investigated — it was decorated.

The ask

A client ran a small electronics resale business. Someone on a marketplace had been buying returned stock in bulk under the seller name `voltbin_ua`, then reselling it as new and, when buyers complained, posting the client's real company address as the "warehouse" for angry pickups. The client wanted one thing:

Who is behind `voltbin_ua`, and is it one person or a shopfront?

That is all. Not an arrest. Not a home. A name, and a confidence level honest enough to act on. Everything below serves that one sentence.

I opened a fresh case note with three columns: claim, evidence, rung. Nothing goes in the report until it earns one.

The clue and the first wall

One clue to start: the seller page.

```
marketplace – seller profile
-----
seller:    voltbin_ua
joined:    2023-11
sales:     412
rating:    4.1
location:  Kyiv (self-reported)
contact:   via platform messages only
-----
```


marketplace public seller card

That is the whole public surface. No phone. No email. Messaging locked behind the platform. The front door is shut; the side windows are another matter.

So I stopped trying the front door. I did not attempt to buy an item to trigger a contact reveal — that manufactures a transaction and puts a real person on the hook. Open sources only.

The only thing the wall gave me for free was the string itself: `voltbin_ua`.

Pivoting the handle

So I ran `voltbin_ua`, and the root `voltbin`, across platforms — by hand and with a handle-search tool, then verified every hit by *content*, never by the mere match.

```
handle sweep – root "voltbin"
-----
voltbin_ua  telegram  EXISTS
voltbin     instagram EXISTS
voltbin     github   404
voltbin_ua  twitter/x  suspended
voltbin.shop domain   parked, no records
voltbin     reddit   EXISTS (1 post, 2019)
-----
```

username enumeration, unverified

Six hits. Most are noise until proven otherwise. Watch how fast they fall.

The reddit voltbin — one post, 2019, in an aquarium-keeping forum, written in English from a user whose other comments place them firmly in Canada. Our seller joined the marketplace in 2023 and trades in Ukraine. Wrong topic, wrong country, wrong years. I did not reject it because the name was *slightly* different — it was identical. I rejected it on content and dates.

Rejected: reddit/voltbin. Not our target. Rung: n/a.

The github 404 and the parked domain are simply empty. Empty is not a finding. I wrote "no data" and moved on — an absence is only useful if you first expected a presence.

That leaves telegram and instagram. Both alive. Both worth a slow look.

Telegram — public only

The Telegram handle voltbin_ua resolved to a user. Public profile, no bio, generic avatar. But the handle appeared as an *author* in two public resale groups — open channels anyone can read without joining anything private.

```
telegram — public group mention
-----
group: @kyiv_tech_resale (public)
from: voltbin_ua
text: "partii lcd matrytsi, samovyviz
      Livyi bereh, opt vid 10 sht"
date: 2024-02-18
-----
```

public channel post, machine-read

Ukrainian: *"batches of LCD matrices, self-pickup, Left Bank, wholesale from 10 units."* Same trade, same city, consistent with the marketplace seller. That is a moderate-rung link between the marketplace `voltbin_ua` and the Telegram `voltbin_ua` — the topic, geography and timing align, but a shared handle plus a shared theme is still just two signals pointing the same way. It is not proof one hand types both.

I pulled the numeric Telegram user ID and logged it. Handles change; the ID is the spine of the account. If `voltbin_ua` renames itself tomorrow, the ID still stitches the history together. I did not join any closed group, did not message the account, did not add the number to a contact book to farm a name. Public read, ID recorded, stop.

One quiet thing: the account had forwarded a message once, from a channel called `@denys_zip_kyiv`. A forward is a relationship of *some* kind — but a weak one, and I want to be clear up front about what it is not. It is not a self-link, not a shared photo, not a shared room. I parked it as a loose thread. It comes back late in this chapter, and — spoiler — it never earns its way into the answer.

The face problem

The instagram `voltbin` had eleven photos — mostly boxes of stock, two of a workbench, and one of a man holding a soldering iron, face clear, taken in what looks like a garage.

A face is a bridge, not a name. I did not run it through a face-search engine that scrapes the open web to find every other photo of this human — the client's question did not need that, and that tool reaches past open posting into building a biometric dossier on a private individual. I stopped at that line and wrote down that I stopped, and why.

What I *could* do with open sources: compare the workbench photo to images already attached to accounts I had lawful public reason to view. The Telegram `voltbin_ua` had, months earlier, posted a photo of a repair bench in a public group. Same distinctive green cutting mat, same brand of magnifier lamp, same window bars in the background.

```
visual cross-check (analyst notes)
```

```
-----
```

```
IG  "voltbin" bench photo:
```

- green cutting mat, worn corner
- magnifier lamp, ring type
- barred window, 4 vertical bars

```
TG  bench photo (public group):
```

- green cutting mat, worn corner
- magnifier lamp, ring type
- barred window, 4 vertical bars

```
-----
```

manual scene comparison, not biometric

This is *scene* matching — the room, not the man. Two accounts photographed the same physical workbench. Combined with the shared handle and the shared trade, the marketplace / Telegram / Instagram cluster now holds together at a strong

rung — same *workshop*, near-certainly. Whether that means one operator or two people who share the room is a question the disproof pass has to settle later; the room-match alone can't. Still no name.

The bridge from a leak

To turn this cluster into a *person*, I needed an identifier that lives in the world — an email, a phone — tied to one of these accounts through an open source, not a purchase.

The Instagram account, years back, had linked out to a defunct personal blog, voltbin-repairs.blogspot, archived. The archived contact page listed a business email:

```
web archive — blog contact page
-----
captured: 2022-06-30
"Napyshit: voltbin.fix@gmail.com"
"Telegram: @voltbin_ua"
-----
```

archived page, public capture

An email the account *published itself*. That is fair game — it was posted openly for customers. Does that email appear in any known public breach corpus? A breach can tell you an email *co-occurred* with a name or a phone in some dump. It cannot, by itself, tell you the truth of that pairing. Dumps are dirty. People lie on signup forms. Credentials get stuffed.

```
breach-index lookup (fictional)
-----
query: voltbin.fix@gmail.com
-----
source A  "electro-forum-2021"
  username: voltbin
  reg_name: Denys K.
  phone:    +380 (67) 555 0148
source B  "coupon-site-2020"
  name:     Denys Koval
  city:     Kyiv
source C  "gaming-2019"
  name:     Artem Bond
  phone:    +380 (50) 555 0921
-----
```

aggregated breach index result

Three sources. Two of them (A, B) agree on a first name: Denys, one gives a surname Koval, Kyiv, and a phone. The third (C) says Artem Bond with a different phone. That contradiction is not a nuisance — it is information. One of these is our operator's real signup, and at least one is noise: a reused password stuffed elsewhere, or someone who typed this email by accident, or a deliberately fake name.

I did not pick "Denys" because two beats one and it felt right. Two dirty sources can share the same error. I marked the whole email→name step as weak until something *outside* the breach corpus backed it. This is the moment most cases go wrong — a plausible name appears and the investigator quietly promotes a guess to a fact.

The circle, and the locked door

The Family Chain move. When a target's own profile is shut, the people around them often are not, and the relationships are load-bearing.

First, a temptation I refused. The dump had *whispered* a surname — "Koval," from source B — and the parked forward came from a channel called @denys_zip_kyiv, a name-root that also looked like "Denys." It would have been easy to braid those two threads together and announce "Denys Koval." But look at what that braid is made of: a dirty breach record and a handle-string. Two weak, self-set-adjacent scraps do not become strong by being tied to each other. I set both aside and went looking for the surname somewhere that could actually *carry* it.

I found it in the account cluster itself. The Instagram voltbin had one photo it had been publicly **tagged in** by a different account — @koval_secondhand, a small second-hand electronics shop. A tag is a two-way public act: the shop chose to attach itself to our operator's post, in the open, years before I arrived. That link does not route through @denys_zip_kyiv at all. It stands on its own.

I looked at @koval_secondhand publicly, and the room did the rest. One of its bench photos showed the *same green cutting mat, worn at the same corner, and the same four-barred window* already tied to the cluster by the scene match. So this account

was not a stranger who happened to share a name — it was photographed inside the very workshop the whole cluster keys off. And it was run, openly, by one **O. Koval**.

The breach's earlier "Koval" whisper now reads as corroboration — a third road that happens to agree — never the source.

I did try to reach the operator's own name head-on, and hit a wall: the personal Facebook implied by the phone in source A was fully locked. No public posts, no friend list, nothing. A dead end by the front door — again.

So, the Chapter Six move: not *through* the locked door, *around* it. @koval_secondhand was public and chatty. A post there, wishing someone a birthday, named "Denys" and showed a photo at a workbench — the *same green cutting mat and barred window* from our scene match.

```
public post – @koval_secondhand
-----
date: 2024-05-12
text: "Z dnem narodzhennia, brat!
      Denys, nekhai remontuietsia
      lehko :)"
photo: bench, green mat, barred
       window (matches cluster)
-----
```

relative's public birthday post

A relative, openly, put a first name on the exact room we had already tied to the account cluster by an independent method. The name Denys was no longer arriving for free — it was

arriving from the *scene* now, not the dump.

I did not contact the relative or ask them anything. I read what was already public and stopped — pretexting a family member to coax a surname is over the line.

Stacking, then trying to break it

Lay the genuinely independent factors side by side for the `voltbin_ua` → Denys Koval claim. Two facts arriving through one account are one source, not two votes — that is the trap this list is arranged around:

- Shared handle across marketplace, Telegram, Instagram (*factor 1*)
- Same physical workbench in TG + IG photos (*factor 2 — a room, not a string, so independent of the handle*)
- Self-published email on archived blog (*factor 3*)
- The name — surname *Koval* and first name *Denys* both arriving via `@koval_secondhand`, one account roped to the cluster by the room-match. Two facts, one source (*factor 4*)

So the honest count is three-to-four independent factors, not five. Absent by design: the breach dump's name-guess and the `@denys_zip_kyiv` forward — neither carries the identity. Factor 4 is a single point of failure for the *name*; factors 1–3 hold the single-operator finding on their own. That layered structure moves the claim to **strong**, and, with the disproof pass below

surviving, to strong-leaning — one signal short of confirmed while the brother rival stays open.

Then I did the thing that feels like sabotage and isn't: I tried to *disprove* it. If I were wrong, what would I expect to see?

- If `voltbin_ua` were a *shopfront* — several people sharing one login — the writing style across the Telegram posts should drift: different hours, different phrasing, different dialect. It didn't. Consistent evening postings, consistent Surzhyk (mixed Ukrainian-Russian speech) phrasing, consistent typo (*samovyviz* always one word). Reads as one hand.
- If the *email* belonged to someone else who merely reused it, the self-published blog linking that email to `@voltbin_ua` in 2022 would have to be a coincidence three ways over. It survives the challenge.
- If "Denys Koval" were a stitch of two different people, the workbench room would not appear under both the account cluster *and* the relative's post. It appears under both.
- The hardest rival, and the one the tidy version skips: what if the operator is O. Koval — the brother who runs `@koval_secondhand` — and *Denys* is merely the sibling who also shows up at that bench? Both Kovals use that room, so the room-match cannot on its own tell them apart. Two thin facts lean the operator to Denys: the operator's *own* published identifier — email `voltbin.fix@gmail.com` and handle `voltbin` — co-occurs in the breach corpus with Denys (sources A and

B), never with O.; and O. Koval, in the birthday post, addresses Denys as *brat* — writing *to* the operator identity from outside it, not *as* it. Dirty tie, unclean lean — the rival doesn't fully die. It shrinks to a residual, and I keep it named.

voltbin_ua is a single operator working from the Koval workshop — most likely Denys Koval, Kyiv, with his brother O. Koval the named residual alternative I could not cleanly exclude. Rung: strong, bordering confirmed.

The branch that would not close

Now the part the clean textbooks skip.

The client's fuller worry was that voltbin_ua was the front for an *organised return-fraud ring* — that "Denys" was one node among several coordinated sellers. The forward from @denys_zip_kyiv, the brother's shop, a couple of other resale handles that had cross-posted the same stock photos — it *smelled* like a network.

I chased it. And I have to tell you honestly: I could not close it.

```
network hypothesis – status
-----
voltbin_ua  -> Denys Koval  STRONG
koval_2hand -> O. Koval    related
denys_zip   -> ??? same as Denys?
    handle name matches, but:
    - different posting hours
    - no shared photos
    - no self-link either way
    status: UNRESOLVED
ring coordination?  NOT SHOWN
-----
```

analyst status board, open items

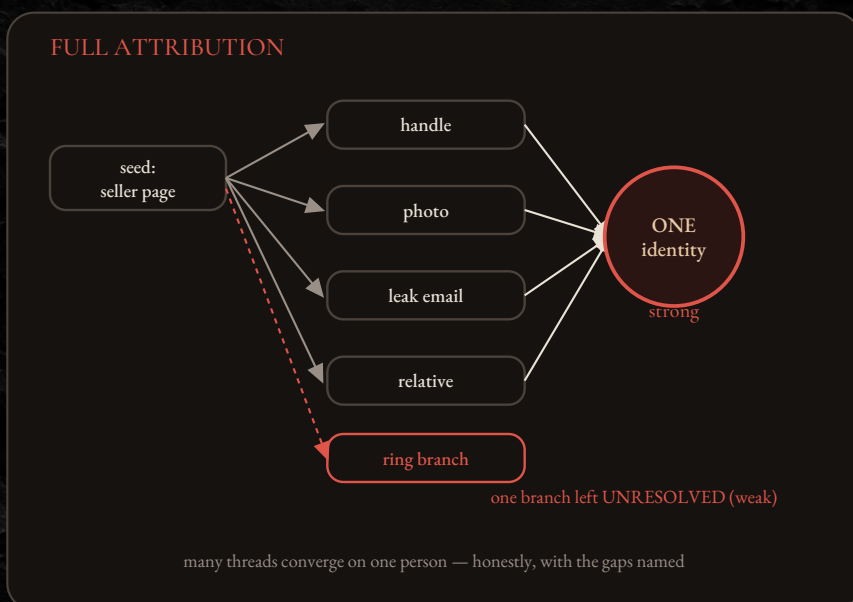
@denys_zip_kyiv shares a first name and a forwarded message with our target. That is it. The posting hours don't match the established pattern. No shared photos. Neither account ever links to the other as *self*. It could be the same Denys running a second account. It could be a genuinely different Denys in the same small trade — and in Kyiv's parts scene, common first name, overlapping suppliers, that is entirely ordinary. A shared handle-root is the weakest signal there is, and a forward is a contact, not an identity.

So I did not resolve it.

And this is the part that keeps the whole report honest: because that branch stayed weak, it fed *nothing* into the attribution above it. Strike @denys_zip_kyiv from the case entirely and "Denys Koval, single operator, strong" does not move a millimetre. That is the quarantine test — if deleting an

unresolved lead would weaken your confident finding, a guess was propping up an answer. Here, it wasn't.

The honest rung here is weak, and weak does not go in the conclusions. It goes in the "could not confirm" section, by name, so the client knows exactly what I don't know.



The whole graph on one page: the voltbin_ua seed, pivoting out across marketplace, Telegram, Instagram and an archived blog, with four independent signals (handle, room-match, self-published email, room-tied relative account) converging on one identity — Denys Koval, strong. The @denys_zip_kyiv branch hangs off to the side, dotted and open: reached but never resolved, and deliberately load-bearing for nothing.

The decision-grade mini-report

Everything above compresses to one page a client can act on.

ATTRIBUTION SUMMARY – voltbin_ua

ASKED:

Who operates seller "voltbin_ua",
and is it one person or a front?

FOUND:

Single operator, working from the
Koval workshop, most likely
Denys Koval, Kyiv, electronics
repair/resale. Named residual:
brother O. Koval (shared bench).

CONFIDENCE:

strong: single operator working
from the Koval workshop
strong-leaning: Denys specifically,
with brother O. Koval as a
named residual

BASIS (independent factors, ~3-4):

- handle reuse across 3 platforms
- shared physical workbench in
TG + IG photos (scene match)
- self-published email, archived
- name (surname + first name) from
one room-tied account
(@koval_secondhand) - counts as
ONE source, not two, not the dump

RULED OUT:

- reddit "voltbin": wrong country,
topic, years (content-rejected)
- "Artem Bond" in breach index:
unsupported outside the dump
- shopfront theory: writing style
reads as one hand, not many

COULD NOT CONFIRM:

- that voltbin_ua is part of an
organised ring

- that @denys_zip_kyiv is the same person (weak; name-root only)
- > DO NOT CONCLUDE either.

NOT DONE (ethical line):

- no face-search, no purchase to unmask, no locked-profile access
- no messaging target or relatives

SOURCES:

marketplace card; 2 public TG groups; IG public posts; archived blog (web capture 2022-06-30); breach index (3 sources, treated as dirty); relative's public post. All captures archived + hashed.

deliverable, one page

Note the shape. The confident claim leads. The rejections are named. The unresolved branch is stated *as* unresolved, not buried. What I chose not to do is listed, because a client deserves to know the edges of the work as much as its middle. And the sources carry an archive note — every public page was captured and hashed at the time, because open pages get deleted, and a finding you cannot re-show is a finding you cannot defend.

What this chapter was really about

Nine techniques, and not one of them was the answer on its own. The handle got me nowhere until content killed the false hit. The face I refused to search — the *room* is what talked. The breach handed me a name I was right not to trust, and the same name arriving a second way, from a relative, is what made it real. And the tidiest thread — the ring — is the one I had to leave hanging.

That is cross-platform attribution.

The skill was never finding the signal. It was weighing it.

What You Can Do Now

— *and what comes next.*

Stop for a second and look at how far you have come.

In Volume 1 you could take one clue and find a person. Now you can take a person who is scattered across a dozen platforms — private here, deleted there, hiding behind a second handle somewhere else — and weave them back into one identity you can actually stand behind. You can go around a wall instead of hitting it. You can tell an independent signal from a coincidence. And you can say, out loud and honestly, *how sure you are* — including the hardest sentence in this craft: "I cannot confirm this."

That last skill is the one that separates you now. Anyone can collect. Attribution — deciding that all these scattered pieces are *one real person* — is where the judgement lives, and where the damage is done when judgement fails. You have spent this whole book learning not to fool yourself. Keep it.

Keep the habits, not the tools

Platforms will change. The exact bot, the exact search, the exact site will be gone in a year or renamed. None of that matters. The *shape* of the work does not change: find what is open around a closed door, pivot with discipline, stack independent signals — separate sources, counted once each — weigh every finding on the ladder, and stop at the line. Learn that, and every new platform is just another door for the same method.

Where to keep sharpening

- **Work yourself, across platforms.** Do the whole thing on your own footprint — find every account, every tag, every leak that ties back to you. Then clean it up. Best exercise there is.
- **OSINT CTFs and verification challenges.** They train cross-platform pivoting and the discipline of proof, safely and for free.
- **Missing-persons search events** (such as Trace Labs) — real, meaningful work, alongside people who have done it for years, inside a firm no-contact rule.
- **The reputable guides** — teams like Bellingcat publish careful, ethical, cross-platform methodology worth returning to.

A bridge to Volume Three

This book was about people across the platforms where they *are*. Volume 3 — **Hard Targets** — is about the places they are *not*: the account they deleted and thought was gone, the photo with no caption that you geolocate from what is in the frame, the domain and the infrastructure behind an operation. It is where OSINT stops being about profiles and starts being about the world.

For now, you have done the thing this book was for. You can follow one person across the whole internet — carefully, honestly, and knowing exactly where to stop.

Keep your notes. Stay curious. Stay kind. I will see you in Volume 3.

— Yehor

Appendix

New terms, the method on one page, and templates you can copy.

A — New words in this volume

- **Access layer** — the open things *around* a private profile (mutual friends, tags, cached copies) that leak what the profile itself hides. You will also hear this called the *ring* around the target, or the *leak layers* — same idea, different names. This book uses access layer as the lead term; treat *ring* and *leak layers* as synonyms for it.
- **Combolist** — a plain list of email:password (or username:password) pairs, usually stitched together from many old breaches and passed around. Useful only to *link* identifiers, never to log in.
- **Confirmation by convergence** — how you actually reach *confirmed*: several independent signals pointing at the same person, with nothing contradicting. No single clue proves it; the agreement does. Sometimes called confirmation by independent signals.
- **Disproof pass** — a deliberate step where you stop building the case and try to *break* it — hunting the one fact that would prove you wrong. If the finding survives, it is stronger. Better it fails here than in a report.

- **Dork (search operator)** — a targeted search using an engine's own operators (site:, filetype:, quotes, minus) to force it to show exactly what you want. Often called a "Google dork."
- **EXIF** — hidden data tucked inside a photo file by the camera or phone: date, time, sometimes GPS location and device. Many platforms strip it; some do not. Always worth checking.
- **Family Chain** — this book's name for reaching a private target through their relatives — a parent, sibling, or spouse whose accounts are open and who tags, mentions, or photographs the person you cannot see directly.
- **FOAF** — "friend of a friend"; reaching a target through the people connected to them.
- **Identity graph** — the map of a person as nodes (accounts, people) and links (same face, shared handle, family), which you grow until it points at one identity.
- **Independent signal** — a clue whose source is separate from your other clues. Two clues from the same source count as one.
- **k-anonymity password check** — a safe way to test whether a password appears in known breaches without ever sending the whole thing. You send only the first few characters of its hash; the service returns every match for that prefix and you finish the check on your own machine. This is how Have I

Been Pwned's Pwned Passwords works.

- **Naming grammar** — the personal, consistent way one person mutates their handle (first+last, a favourite number, leetspeak) so you can predict handles you have not found.
- **Numeric ID** — a platform's internal, permanent number for an account (e.g. a Telegram user ID) that stays the same even when the username and display name change. The durable anchor. Two flavours matter: *sequential* IDs (Telegram) are handed out roughly in order, so a lower number means an older account — that is what lets you estimate an account's age; *snowflake* IDs (Discord, Twitter/X) have the creation timestamp mathematically baked into the number itself, so you can decode the exact creation moment. Telegram IDs are sequential, not snowflakes — don't confuse the two.
- **Page source** — the raw HTML behind a web page (right-click → "View source"). It often holds what the pretty page hides: a numeric ID, a full-size image URL, a real name buried in a tag. *2026 caveat*: on many big platforms the page is now just a JavaScript shell, so the ID and data are loaded after the fact and are not in the raw source anymore — it still pays off on smaller or older sites, and for image URLs.
- **Password fingerprint** — a rare, reused password appearing on two accounts, used to *link* them (never to log in).

- **PYMK (People You May Know)** — a platform's friend-suggestion feature. Because it suggests people from hidden connections, it can quietly reveal who is linked to whom — a pivot, if you read it carefully.
- **Resolver (handle → numeric ID)** — a small tool or service that turns a public username into the account's permanent numeric ID (and back), so you can keep tracking someone after they rename.
- **Stealer log vs breach** — two different kinds of leaked data. A breach is data spilled from one hacked service — one site's users. A stealer log is everything a piece of malware scraped from a single infected device — many sites' logins, cookies, and files from one victim. Different shape, different meaning.
- **Stylometry (light)** — telling authors apart by habits of writing: phrasing, punctuation, emoji, timing.
- **Username enumerator** — a tool that checks one username across dozens or hundreds of sites at once to see where it exists. A fast way to start the username web (Sherlock and Maigret are examples).
- **WHOIS** — the public registration record for a domain name: who registered it, when, and through whom. Note: since GDPR, most personal details are redacted behind a privacy service, so WHOIS today is usually thinner than older guides suggest.

B — A starter toolbox (2026)

This book names almost no tools on purpose — the method is what lasts. But beginners keep asking "yes, but *what do I open?*", so here is a short, honest starting kit, grouped by job. Read this caveat first: tool names, owners, and terms change constantly, and any of these could be renamed, paywalled, or gone by the time you read this. Verify a tool's current status and legality before you lean on it. These are places to *learn the method*, not a crutch to skip it.

- **Breach membership** — Have I Been Pwned (has an account or address turned up in a known breach); its Pwned Passwords service for the safe k-anonymity password check.
- **Username enumeration** — Sherlock, Maigret, WhatsMyName (check one handle across many sites at once).
- **Image and face** — Yandex Images, Google Lens, TinEye for reverse-image search. Dedicated *face*-search tools such as PimEyes exist, but they raise heavy ethics and legal flags (biometrics under GDPR and BIPA) — treat them as a red line to think hard about, not a default.
- **Archiving** — the Wayback Machine and archive.today, both to *retrieve* vanished pages and to *preserve* what you find before it disappears.
- **Frameworks and directories** — the OSINT Framework and IntelTechniques, curated maps of what exists by category.

Every one of these is a starting point for a way of *thinking* you already learned in these chapters. Method first, tools second — a tool is only ever a faster way to do something you could do by hand.

C — The method on one page (cross-platform)

1. Start from one clue on one platform. 2. When you hit a wall, go around it — work the access layers and the circle, not the locked door. 3. Pivot the identifier (handle, email, face, number) across every platform it might reach. 4. Stack independent signals — and only count them once each. 5. Climb the ladder — weak → moderate → strong → confirmed. *Strong* is earned when the independent signals are jointly hard to explain by any rival hypothesis, not when they simply pile up — count is a floor, not the test. 6. Try to break it — hunt the evidence that would prove you wrong before you conclude. 7. Write it down as you go — screenshot, URL, date, and an archived copy. 8. Stop at the line — take only what answers the question; leave the private lives you passed through alone.

D — Where to practise (safely and legally)

- Investigate yourself, across platforms — find every account, tag, and leak that ties back to you, then clean it up.
- OSINT CTFs and verification challenges — built for cross-platform pivoting and proof discipline.

- Missing-persons search events (e.g. Trace Labs) — real, supervised, no-contact.
- Reputable guides (e.g. Bellingcat's toolkit) — careful, ethical method to return to.

You never need to spend a cent, and you never need a real third-party target to learn this.

E — Templates you can copy

The attribution report. Every cross-platform finding you hand to anyone fits this:

```

CASE:          (one line – what this is)
DATE:          (when)
OBJECTIVE:     (the single question, one sentence)

IDENTITY:      (who – name/handle, plainly)
CONFIDENCE:    (rung: weak/moderate/strong/confirmed
               + the independent signals it rests on)
RULED OUT:     (look-alikes checked and rejected, why)
NOT DONE:      (limits, the line you stopped at,
               what you could NOT confirm)
SOURCES:       (numbered: platform + URL + date +
               archived copy – include dead ends)
  
```

The signal table. Fill one row per clue; the last column is the one that matters:

SIGNAL	VALUE	INDEPENDENT?
same handle	mossyfern	weak (common)
profile photo	matches	yes
posting hours	20:00-23:00	yes (own signal)
self-set name	"Deb"	NO (same source)

Caveat: observed style/timing signals from the same city or scene are correlated — collapse them into one factor, weighted by rarity, before counting. A shared "posting hours" is not a free independent vote if two same-population strangers would share it.

The identity graph (sketch). Nodes are accounts and people; links are what ties them. Grow it until it points at one person:

```
TARGET (locked)
| [wall]
+--> WIFE (open) --> SISTER (open)
                        |
                        +--> family photo
                           tags TARGET's
                           2nd handle
```

F — Capture it, or it didn't happen

A finding you cannot re-show is a finding you cannot defend. Sources vanish, profiles go private, posts get deleted — often the moment they matter. So for anything load-bearing — any fact your conclusion rests on — preserve it at the moment you see it, not later.

For each such source, record:

- The exact URL — the full link, not "I found it on their profile."
- The date and time you saw it, in UTC — one clock everyone can compare, no timezone guessing.

- ****A screenshot *and* an independent archive**** — save it to the Wayback Machine *and* to archive.today. Two archives so one covers the other if a service is down, blocked, or later purged.
- **A hash of any file you download** — a SHA-256 of the image, PDF, or dump. It is a fingerprint: anyone can re-hash the file later and prove it is the exact bytes you saw, unchanged.
- ****When the source was *created* vs when you *saw* it**** — a post written in 2019 and read by you in 2026 are two different dates. Note both; don't let "seen" masquerade as "created."

Keep a running source log — one line per source, in the order you found them:

```
URL | seen-UTC | archive | hash | note
```

the skeleton — one field per column

```
URL      blog.example/about
seen-UTC  2026-05-12T14:03Z
archive   wayback+today
hash      sha256 ab12c...
note      about page
```

a filled row from a real capture log

One row per load-bearing source; fill it as you go, not at the end.

If you cannot name the confidence rung, you have not finished the finding.

G — Red lines (one card)

Scattered through the chapters, one page. These are professional declines, not moral lectures — the answers you give the client, or yourself, when the tempting move crosses a line.

- Never type a leaked password into a login. Passwords from breach corpora are a *linking* signal, never a key. Using one is unauthorized access. (Ch5)
- Never run face-search on non-consenting people in jurisdictions where it's a legal grey line. Scene-match the room, not the face. Biometric scraping (GDPR, BIPA) is a red line to think hard about, not a default. (Ch4)
- Never chase a target's relatives to locate someone hiding for their safety. The Family Chain is a real technique and a real hazard; when the client's interest is finding a person who left them, decline the job. (Ch6)
- Never unmask an anonymous source, protester, or whistleblower for a client whose interest is retaliation. Anonymity is the safety layer; stripping it for a hostile requester is the harm. (Ch3)
- Assume every third-party bot or lookup service logs your query. The operator sees what you asked. Treat "free intel tool" as a honeypot until proven otherwise. (Ch3)
- Never pretext, deceive, or social-engineer your way in. No fake identity to a relative, no manufactured transaction to

unmask a seller, no cold DM to coax a name. Open sources only. (Ch1, Ch9)